

1

Tensor Algebra

Underlying everything we study is the field of real numbers $\mathbb{R}$ and three-dimensional Euclidean space $\mathbb{E}^3$. We refer to elements of $\mathbb{R}$ as **scalars** and denote them by light-face symbols such as a , b , x and y . We refer to elements of $\mathbb{E}^3$ as **points** and denote them by bold-face symbols such as $\mathbf{a}$, $\mathbf{b}$, $\mathbf{x}$ and $\mathbf{y}$.

The aim of this chapter is to build up a mathematical structure for $\mathbb{E}^3$ and various other sets associated with it, such as the set of vectors $\mathcal{V}$. The important ideas that we introduce are: (i) the distinction between points and vectors; (ii) the distinction between a vector and its representation in a coordinate frame; (iii) index notation; (iv) the set of second-order tensors $\mathcal{V}^2$ as linear transformations in $\mathcal{V}$; (v) the set of fourth-order tensors $\mathcal{V}^4$ as linear transformations in $\mathcal{V}^2$.

1.1 Vectors

By a **vector** we mean a quantity with a specified magnitude and direction in three-dimensional space. For example, a line segment in space can be interpreted as a vector as can other things such as forces, velocities and accelerations. We denote a vector by a bold-face symbol such as $\mathbf{v}$ and denote its magnitude by $|\mathbf{v}|$. Notice that for any vector $\mathbf{v}$ we have $|\mathbf{v}| \geq 0$. It is useful to define a **zero vector**, $\mathbf{0}$, as a vector with zero magnitude and no specific direction. Any vector having unit magnitude is called a **unit vector**.

A vector may be represented graphically by an arrow as depicted in Figure 1.1. The orientation of the arrow represents the direction of the vector and the length of the arrow represents the magnitude. We will always identify vectors with arrows in this way. Moreover, two vectors

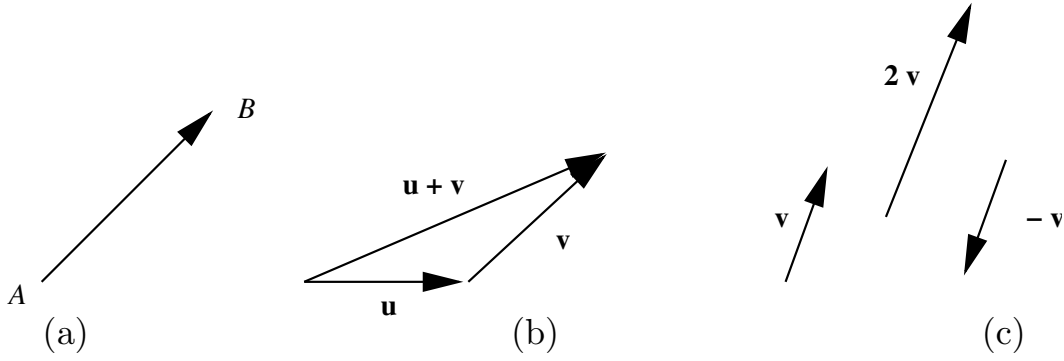


Fig. 1.1 Graphical illustration of a vector and basic vector operations. (a) the tail A and tip B of a vector. (b) the sum of two vectors. (c) the multiplication of a vector by a scalar.

will be considered equal if they have the same direction and magnitude, regardless of their location in space.

Scalars and vectors are examples of more general objects called **tensors**. For example, a scalar is a zeroth-order tensor, while a vector is a first-order tensor. Later we will consider second-order and fourth-order tensors, but for now we concentrate on first-order ones, that is, vectors.

1.1.1 Vector Algebra

If we let $\mathcal{V}$ be the set of all vectors, then $\mathcal{V}$ has the structure of a real vector space since

$$\mathbf{u} + \mathbf{v} \in \mathcal{V} \quad \forall \mathbf{u}, \mathbf{v} \in \mathcal{V} \quad \text{and} \quad \alpha \mathbf{v} \in \mathcal{V} \quad \forall \alpha \in \mathbb{R}, \mathbf{v} \in \mathcal{V}.$$

In particular, we define the sum $\mathbf{u} + \mathbf{v}$ as that element of $\mathcal{V}$ which completes the triangle when $\mathbf{u}$ and $\mathbf{v}$ are placed tip-to-tail as shown in Figure 1.1(b). Given any scalar $\alpha \in \mathbb{R}$ we define the product $\alpha \mathbf{v}$ as that element in $\mathcal{V}$ with magnitude $|\alpha||\mathbf{v}|$, where $|\alpha|$ denotes the absolute value of α , and direction the same as or opposite to $\mathbf{v}$ depending on whether α is positive or negative, respectively, as depicted in Figure 1.1(c). If $\alpha = 0$, then $\alpha \mathbf{v}$ is the zero vector.

With the vector space $\mathcal{V}$ in hand we can define some useful mathematical operations between elements of $\mathbb{E}^3$ and $\mathcal{V}$. Given any two points $\mathbf{x}$ and $\mathbf{y}$ we define their difference $\mathbf{y} - \mathbf{x}$ to be the unique vector which points from $\mathbf{x}$ to $\mathbf{y}$, as shown in Figure 1.2(a). Hence given $\mathbf{x}, \mathbf{y} \in \mathbb{E}^3$ we have $\mathbf{y} - \mathbf{x} \in \mathcal{V}$. Given any point $\mathbf{z}$ and any vector $\mathbf{v}$ we define the sum $\mathbf{z} + \mathbf{v}$ to be the unique point such that $(\mathbf{z} + \mathbf{v}) - \mathbf{z} = \mathbf{v}$, as shown in Figure 1.2(b). Hence given $\mathbf{z} \in \mathbb{E}^3$ and $\mathbf{v} \in \mathcal{V}$ we have $\mathbf{z} + \mathbf{v} \in \mathbb{E}^3$.

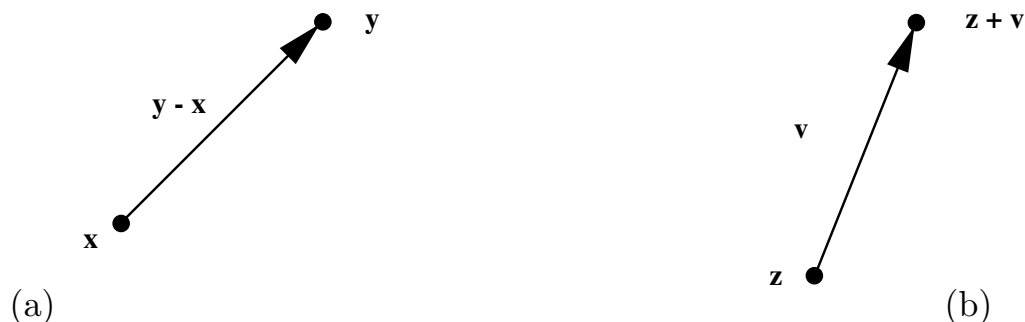


Fig. 1.2 Graphical interpretation of operations between points and vectors. (a) the difference between two points is a vector. (b) the sum of a point and a vector is a point.

We next define some geometrical operations between vectors which will allow us to construct mathematical representations for $\mathcal{V}$ and $\mathbb{E}^3$.

1.1.2 Scalar and Vector Products

The **scalar** or **dot product** of two vectors $\mathbf{a}$ and $\mathbf{b}$ is defined geometrically as

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta,$$

where $\theta \in [0, \pi]$ is the angle between the tips of $\mathbf{a}$ and $\mathbf{b}$. Two vectors $\mathbf{a}$ and $\mathbf{b}$ are said to be **orthogonal** or **perpendicular** if $\mathbf{a} \cdot \mathbf{b} = 0$. For any vector $\mathbf{a}$ notice that $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

The **vector** or **cross product** of two vectors $\mathbf{a}$ and $\mathbf{b}$ is defined geometrically as

$$\mathbf{a} \times \mathbf{b} = (|\mathbf{a}||\mathbf{b}| \sin \theta) \mathbf{e},$$

where, as before, $\theta \in [0, \pi]$ is the angle between the tips of $\mathbf{a}$ and $\mathbf{b}$, and $\mathbf{e}$ is a unit vector perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$. The orientation of $\mathbf{e}$ is defined such that a right-handed rotation about $\mathbf{e}$, through an angle θ , carries $\mathbf{a}$ to $\mathbf{b}$. If $\mathbf{a} \times \mathbf{b} = \mathbf{0}$, then $\mathbf{a}$ and $\mathbf{b}$ are said to be **parallel**. Notice that the magnitude of $\mathbf{a} \times \mathbf{b}$ is the same as the area of a parallelogram with sides defined by $\mathbf{a}$ and $\mathbf{b}$.

Example: Let $V = |\mathbf{a} \times \mathbf{b} \cdot \mathbf{c}|$. Then V is the volume of the parallelepiped defined by $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. In particular, we have

$$V = |\mathbf{a}||\mathbf{b}||\mathbf{c}| \sin \theta \cos \phi,$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and ϕ is the angle between $\mathbf{c}$ and

$\mathbf{a} \times \mathbf{b}$. The factor $|\mathbf{a}||\mathbf{b}|\sin\theta$ is the area of the parallelogram defined by $\mathbf{a}$ and $\mathbf{b}$, and the factor $|\mathbf{c}|\cos\phi$ gives the height of the parallelepiped with respect to the plane of $\mathbf{a}$ and $\mathbf{b}$. $\square$

1.1.3 Projections, Bases and Coordinate Frames

Let $\mathbf{e}$ be a unit vector. Then any vector $\mathbf{v}$ can be written in the form

$$\mathbf{v} = \mathbf{v}_e + \mathbf{v}_e^\perp, \quad (1.1)$$

where $\mathbf{v}_e$ is parallel to $\mathbf{e}$ and $\mathbf{v}_e^\perp$ is perpendicular to $\mathbf{e}$. The vector $\mathbf{v}_e$ is called the **projection** of $\mathbf{v}$ parallel to $\mathbf{e}$ and is defined as

$$\mathbf{v}_e = (\mathbf{v} \cdot \mathbf{e})\mathbf{e}.$$

The vector $\mathbf{v}_e^\perp$ is called the projection of $\mathbf{v}$ perpendicular to $\mathbf{e}$. Since $\mathbf{v}_e^\perp = \mathbf{v} - \mathbf{v}_e$ we find that

$$\mathbf{v}_e^\perp = \mathbf{v} - (\mathbf{v} \cdot \mathbf{e})\mathbf{e}.$$

Thus any vector $\mathbf{v}$ can be uniquely decomposed into parts parallel and perpendicular to a given unit vector $\mathbf{e}$.

By a right-handed orthonormal **basis** for $\mathcal{V}$ we mean three mutually perpendicular unit vectors $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ which are oriented in the sense that

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}.$$

While any three mutually perpendicular unit vectors necessarily satisfy $|(\mathbf{i} \times \mathbf{j}) \cdot \mathbf{k}| = 1$, a right-handed basis has the property that $(\mathbf{i} \times \mathbf{j}) \cdot \mathbf{k} = 1$. In contrast, a left-handed basis satisfies $(\mathbf{i} \times \mathbf{j}) \cdot \mathbf{k} = -1$.

Using the notation $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ in place of $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ we find, by repeated application of (1.1), that any vector $\mathbf{v}$ can be uniquely decomposed as

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3 \quad \text{where} \quad v_i = \mathbf{v} \cdot \mathbf{e}_i \in \mathbb{R}.$$

The numbers v_1, v_2, v_3 are called the **components** of $\mathbf{v}$ in the given basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. For brevity we will often denote the set of components by v_i and a given basis by $\{\mathbf{e}_i\}$, where it is understood that the subscript i ranges from 1 to 3.

We will frequently find it useful to arrange the components of a vector $\mathbf{v}$ into a 3×1 column matrix $[\mathbf{v}]$, in particular

$$[\mathbf{v}] = \begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} \in \mathbb{R}^3,$$

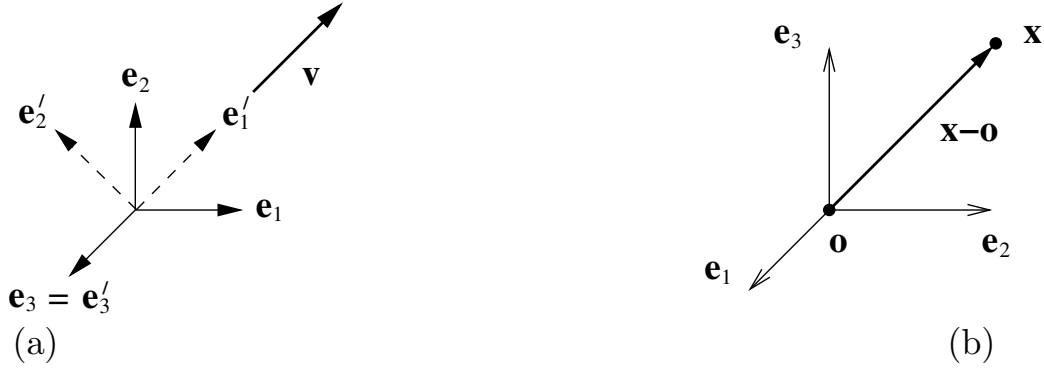


Fig. 1.3 Graphical illustration of orthonormal bases and a coordinate frame. (a) two orthonormal bases for $\mathcal{V}$. (b) Cartesian coordinate frame for $\mathbb{E}^3$.

where $\mathbb{R}^3$ denotes the set of all ordered triplets of real numbers. We will also have need to consider the **transpose** of $[\mathbf{v}]$, which is a 1×3 row matrix defined as $[\mathbf{v}]^T = (v_1, v_2, v_3)$. Notice that by properties of the transpose we may write $[\mathbf{v}] = (v_1, v_2, v_3)^T$. We call $[\mathbf{v}]$ the **matrix representation** of $\mathbf{v}$ in the given basis.

It is important to distinguish between vectors $\mathbf{v} \in \mathcal{V}$ (arrows in space) and their representations $[\mathbf{v}] \in \mathbb{R}^3$ (triplets of numbers). In particular, a given vector can have different representations depending on the choice of basis for $\mathcal{V}$.

Example: Let $\{\mathbf{e}_i\}$ be a basis for $\mathcal{V}$ and consider a vector $\mathbf{v}$ whose representation in this basis is $[\mathbf{v}] = (1, 1, 0)^T$ as shown in Figure 1.3(a). Thus

$$\mathbf{v} \cdot \mathbf{e}_1 = 1, \mathbf{v} \cdot \mathbf{e}_2 = 1, \mathbf{v} \cdot \mathbf{e}_3 = 0 \quad \text{or} \quad \mathbf{v} = 1\mathbf{e}_1 + 1\mathbf{e}_2 + 0\mathbf{e}_3.$$

Next, consider the same vector $\mathbf{v}$, but rotate the basis $\{\mathbf{e}_i\}$ through an angle of $\pi/4$ about the axis defined by $\mathbf{e}_3$. The result of this rotation is a new basis $\{\mathbf{e}'_i\}$, and in this basis the components of $\mathbf{v}$ are

$$\mathbf{v} \cdot \mathbf{e}'_1 = \sqrt{2}, \mathbf{v} \cdot \mathbf{e}'_2 = 0, \mathbf{v} \cdot \mathbf{e}'_3 = 0 \quad \text{or} \quad \mathbf{v} = \sqrt{2}\mathbf{e}'_1 + 0\mathbf{e}'_2 + 0\mathbf{e}'_3.$$

The representation of $\mathbf{v}$ in the new basis is $[\mathbf{v}]' = (\sqrt{2}, 0, 0)^T$ and we see that $[\mathbf{v}]' \neq [\mathbf{v}]$. $\square$

By a Cartesian **coordinate frame** for $\mathbb{E}^3$ we mean a reference point $\mathbf{o} \in \mathbb{E}^3$ called an **origin** together with a right-handed orthonormal basis $\{\mathbf{e}_i\}$ for the associated vector space $\mathcal{V}$ as depicted in Figure 1.3(b). To any point $\mathbf{x} \in \mathbb{E}^3$ we then ascribe **coordinates** x_i according to the expression

$$x_i = (\mathbf{x} - \mathbf{o}) \cdot \mathbf{e}_i.$$

Thus the coordinates of $\mathbf{x}$ are the unique numbers x_i such that

$$\mathbf{x} - \mathbf{o} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3.$$

Throughout our developments we will use the term coordinate frame for $\mathbb{E}^3$ without making explicit reference to the point $\mathbf{o} \in \mathbb{E}^3$ chosen as the origin. Moreover, we will frequently identify a point $\mathbf{x} \in \mathbb{E}^3$ with its position vector $\mathbf{x} - \mathbf{o} \in \mathcal{V}$ and use the same symbol $\mathbf{x}$ for both.

1.2 Index Notation

In this section we express various operations between vectors in terms of their components in a given frame. We begin with the summation convention for index notation and then introduce the Kronecker delta and permutation symbols which will be used throughout the remainder of our developments.

1.2.1 Summation Convention

The representation of vectors and vector operations in terms of components naturally involves sums. For example, let $\mathbf{a}$ and $\mathbf{b}$ be vectors with components a_i and b_i in a frame $\{\mathbf{e}_i\}$. Then

$$\begin{aligned} \mathbf{a} &= a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3 = \sum_{i=1}^3 a_i \mathbf{e}_i, \\ \mathbf{b} &= b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2 + b_3 \mathbf{e}_3 = \sum_{j=1}^3 b_j \mathbf{e}_j, \\ \mathbf{a} \cdot \mathbf{b} &= \left(\sum_{i=1}^3 a_i \mathbf{e}_i \right) \cdot \left(\sum_{j=1}^3 b_j \mathbf{e}_j \right) = \sum_{i=1}^3 \sum_{j=1}^3 a_i b_j \mathbf{e}_i \cdot \mathbf{e}_j = \sum_{i=1}^3 a_i b_i, \end{aligned}$$

where the last line follows from the fact that $\mathbf{e}_i \cdot \mathbf{e}_j = 1$ or 0 depending on whether $i = j$ or $i \neq j$.

The expressions for $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{a} \cdot \mathbf{b}$ each involves a sum over a pair of indices. Because sums of this type occur so often we adopt a convention and abbreviate the above three equations as

$$\mathbf{a} = a_i \mathbf{e}_i, \quad \mathbf{b} = b_j \mathbf{e}_j \quad \text{and} \quad \mathbf{a} \cdot \mathbf{b} = a_i b_i.$$

The convention is: *whenever an index occurs twice in a term a sum is implied over that index.* In particular, $a_i \mathbf{e}_i$ is shorthand for the sum $\sum_{i=1}^3 a_i \mathbf{e}_i$. Unless mentioned otherwise we will always assume that the

summation convention is in force. In this case any pair of repeated indices in a term represents a sum, for example

$$\begin{aligned} a_2 b_j b_j b_3 &= a_2 (b_1 b_1 + b_2 b_2 + b_3 b_3) b_3, \\ a_i b_i + b_i b_i &= (a_1 b_1 + a_2 b_2 + a_3 b_3) + (b_1 b_1 + b_2 b_2 + b_3 b_3), \\ a_i a_j b_j a_i &= a_i a_i a_j b_j = (a_1 a_1 + a_2 a_2 + a_3 a_3)(a_1 b_1 + a_2 b_2 + a_3 b_3). \end{aligned}$$

Any repeated index, such as i or j above, which is used to represent a sum is called a **dummy index**. This terminology reflects the fact that the actual symbol used for a repeated index is immaterial since

$$a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3 = a_j b_j.$$

Notice that the summation convention applies only to *pairs* of repeated indices. In particular, no sums will be implied in expressions of the form a_i , $a_i b_i b_i$, and so on.

Any index which is not a dummy index is called a **free index**. For example, in the equation

$$a_i = c_j b_j b_i,$$

the index i is a free index, while j is a dummy index. Free indices can take any of the values 1, 2 or 3, and are used to abbreviate groups of similar equations. For example, the above equation is shorthand for the three equations

$$a_1 = c_j b_j b_1, \quad a_2 = c_j b_j b_2, \quad a_3 = c_j b_j b_3.$$

Similarly, the equation

$$a_i b_j = c_i c_k c_k d_j$$

is shorthand for nine equations since both i and j are free indices. The nine equations are

$$a_1 b_1 = c_1 c_k c_k d_1, \quad a_1 b_2 = c_1 c_k c_k d_2, \quad \dots \quad a_3 b_3 = c_3 c_k c_k d_3.$$

Notice that each term in an equation should have the same free indices, and the same symbol cannot be used for both a dummy and free index. For example, equations of the form

$$a_i = b_j, \quad a_i b_j = c_i d_j d_j \quad \text{and} \quad a_i b_j = c_i c_k d_k d_j + d_p c_l c_l d_q$$

are all ambiguous and are not permissible in our use of the summation convention.

1.2.2 Kronecker Delta and Permutation Symbols

To any frame $\{\mathbf{e}_i\}$ we associate a **Kronecker delta** symbol δ_{ij} defined by

$$\delta_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases}$$

and a **permutation symbol** ϵ_{ijk} defined by

$$\epsilon_{ijk} = (\mathbf{e}_i \times \mathbf{e}_j) \cdot \mathbf{e}_k = \begin{cases} 1, & \text{if } ijk = 123, 231 \text{ or } 312, \\ -1, & \text{if } ijk = 321, 213 \text{ or } 132, \\ 0, & \text{otherwise (repeated index).} \end{cases}$$

Notice that the numerical values of δ_{ij} and ϵ_{ijk} are the same for any right-handed orthonormal frame.

The Kronecker delta and permutation symbols enjoy certain symmetry properties under index transposition and permutation. In particular, we find that δ_{ij} is invariant under transposition of indices in the sense that $\delta_{ij} = \delta_{ji}$, whereas ϵ_{ijk} changes sign under (pairwise) transposition, namely, $\epsilon_{ijk} = -\epsilon_{jik}$, $\epsilon_{ijk} = -\epsilon_{ikj}$ and $\epsilon_{ijk} = -\epsilon_{kji}$. However, ϵ_{ijk} is invariant under circular or cyclic permutation of indices in the sense that $\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij}$.

The permutation symbol can alternatively be defined as

$$\epsilon_{ijk} = \det([\mathbf{e}_i], [\mathbf{e}_j], [\mathbf{e}_k]),$$

where $([\mathbf{e}_i], [\mathbf{e}_j], [\mathbf{e}_k])$ is the 3×3 matrix with columns $[\mathbf{e}_i]$, $[\mathbf{e}_j]$ and $[\mathbf{e}_k]$. In particular, all the properties outlined above for ϵ_{ijk} under index transposition and permutation can be deduced from properties of the determinant.

1.2.3 Frame Identities

Various useful identities can be established between the vectors of a frame $\{\mathbf{e}_i\}$ and the Kronecker delta δ_{ij} and the permutation symbol ϵ_{ijk} . In particular, from the definition of δ_{ij} we deduce the identity

$$\mathbf{e}_i = \delta_{ij} \mathbf{e}_j.$$

That is, $\delta_{1j} \mathbf{e}_j = \mathbf{e}_1$, $\delta_{2j} \mathbf{e}_j = \mathbf{e}_2$ and $\delta_{3j} \mathbf{e}_j = \mathbf{e}_3$. This identity shows that, when δ_{ij} is summed with another quantity, the net effect is a “transfer of index”. We call this the **transfer property** of δ_{ij} .

From the definition of ϵ_{ijk} we deduce the identity

$$\mathbf{e}_i \times \mathbf{e}_j = \epsilon_{ijk} \mathbf{e}_k,$$

which provides a concise way to express the nine possible vector products between the vectors of a frame, that is, $\mathbf{e}_1 \times \mathbf{e}_1 = \mathbf{0}$, $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$ and so on. A similar identity which can easily be verified is

$$\mathbf{e}_i = \frac{1}{2} \epsilon_{ijk} \mathbf{e}_j \times \mathbf{e}_k.$$

1.2.4 Vector Operations in Components

The Kronecker delta and permutation symbols naturally arise when various operations between vectors are expressed in terms of their components in a given frame.

Scalar product. Let $\mathbf{a} = a_i \mathbf{e}_i$ and $\mathbf{b} = b_j \mathbf{e}_j$. Then

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (a_i \mathbf{e}_i) \cdot (b_j \mathbf{e}_j) \\ &= a_i b_j \mathbf{e}_i \cdot \mathbf{e}_j \\ &= a_i b_j \delta_{ij} \\ &= a_i b_i, \end{aligned}$$

where the last line follows from the transfer property of δ_{ij} , that is, $b_j \delta_{ij} = b_i$. Thus, from the geometrical definition of the scalar product we find $|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a} = a_i a_i$.

Vector product. Let $\mathbf{a} = a_i \mathbf{e}_i$, $\mathbf{b} = b_j \mathbf{e}_j$ and $\mathbf{d} = d_k \mathbf{e}_k$. Then

$$\mathbf{a} \times \mathbf{b} = a_i b_j \mathbf{e}_i \times \mathbf{e}_j = a_i b_j \epsilon_{ijk} \mathbf{e}_k,$$

where the last equality follows from the frame identity $\mathbf{e}_i \times \mathbf{e}_j = \epsilon_{ijk} \mathbf{e}_k$. Thus, if we let $\mathbf{d} = \mathbf{a} \times \mathbf{b}$, then $d_k = a_i b_j \epsilon_{ijk}$.

Triple scalar product. Let $\mathbf{a} = a_i \mathbf{e}_i$, $\mathbf{b} = b_j \mathbf{e}_j$ and $\mathbf{c} = c_m \mathbf{e}_m$. Then

$$\begin{aligned} (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= (a_i b_j \epsilon_{ijk} \mathbf{e}_k) \cdot (c_m \mathbf{e}_m) \\ &= \epsilon_{ijk} a_i b_j c_m \delta_{km} \\ &= \epsilon_{ijk} a_i b_j c_k. \end{aligned} \tag{1.2}$$

Moreover, by properties of the permutation symbol under cyclic permutation of its indices we find

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a} = (\mathbf{c} \times \mathbf{a}) \cdot \mathbf{b}.$$

From elementary vector analysis we recall that

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \det([\mathbf{a}], [\mathbf{b}], [\mathbf{c}]), \tag{1.3}$$

where $([\mathbf{a}], [\mathbf{b}], [\mathbf{c}])$ is the 3×3 matrix with columns $[\mathbf{a}]$, $[\mathbf{b}]$ and $[\mathbf{c}]$. When (1.3) is combined with (1.2) we obtain

$$\det([\mathbf{a}], [\mathbf{b}], [\mathbf{c}]) = \epsilon_{ijk} a_i b_j c_k, \quad (1.4)$$

which provides an explicit expression for the determinant in terms of a triple sum.

Triple vector product. Let $\mathbf{a} = a_q \mathbf{e}_q$, $\mathbf{b} = b_i \mathbf{e}_i$, $\mathbf{c} = c_j \mathbf{e}_j$ and $\mathbf{d} = d_p \mathbf{e}_p$. Then

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= (a_q \mathbf{e}_q) \times (b_i c_j \epsilon_{ijk} \mathbf{e}_k) \\ &= \epsilon_{ijk} a_q b_i c_j \mathbf{e}_q \times \mathbf{e}_k \\ &= \epsilon_{qkp} \epsilon_{ijk} a_q b_i c_j \mathbf{e}_p \\ &= \epsilon_{pqk} \epsilon_{ijk} a_q b_i c_j \mathbf{e}_p, \end{aligned} \quad (1.5)$$

where the last line follows from the fact that $\epsilon_{qkp} = \epsilon_{pqk}$. Thus, if we let $\mathbf{d} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$, then $d_p = \epsilon_{pqk} \epsilon_{ijk} a_q b_i c_j$. Below we show that this expression can be used to derive a fundamental identity for the triple vector product.

Remark: The geometrical definitions of the scalar and triple scalar products imply that these quantities are frame-independent. That is, the scalar and triple scalar products can be computed using components in any coordinate frame, with the same value obtained in each frame. Identifying scalar quantities with this property will be an important theme in the sequel. For other examples see the discussion of the trace, determinant and principal invariants of second-order tensors. $\square$

1.2.5 Epsilon-Delta Identities

By virtue of their definitions in terms of a right-handed orthonormal frame, the permutation symbol and the Kronecker delta satisfy the following identities.

Result 1.1 Epsilon-Delta Identities. Let ϵ_{ijk} be the permutation symbol and δ_{ij} the Kronecker delta. Then

$$\epsilon_{pqk} \epsilon_{nrs} = \delta_{pn} \delta_{qr} - \delta_{pr} \delta_{qn} \quad \text{and} \quad \epsilon_{pqk} \epsilon_{rqs} = 2\delta_{pr}.$$

$\square$

Proof See Exercise 18. $\square$

As an application of the above result we establish an identity involving the triple vector product

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}, \quad \forall \mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathcal{V}.$$

To see this notice that, by (1.5) and Result 1.1, we have

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \epsilon_{pqk} \epsilon_{ijk} a_q b_i c_j \mathbf{e}_p \\ &= (\delta_{pi} \delta_{qj} - \delta_{pj} \delta_{qi}) a_q b_i c_j \mathbf{e}_p \\ &= (a_q c_q) b_p \mathbf{e}_p - (a_q b_q) c_p \mathbf{e}_p \\ &= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}, \end{aligned}$$

which establishes the result.

1.3 Second-Order Tensors

Many physical quantities in mechanics, such as forces, velocities and accelerations, are well-represented by the notion of a vector (first-order tensor). In any Cartesian coordinate frame for $\mathbb{E}^3$ such quantities are described by three components. In our studies we will encounter other physical quantities, such as stress and strain, which are not well-represented by vectors, but rather by *linear transformations between vectors*, leading to the notion of a second-order tensor as defined below. As we will see, such quantities will be described by nine components in any Cartesian coordinate frame for $\mathbb{E}^3$.

1.3.1 Definition

By a **second-order tensor** $\mathbf{T}$ on the vector space $\mathcal{V}$ we mean a mapping $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{V}$ which is *linear* in the sense that:

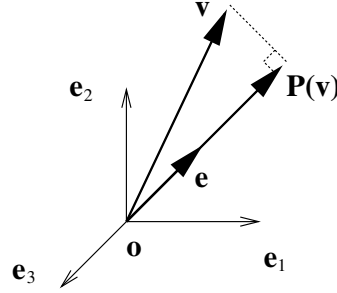
- (1) $\mathbf{T}(\mathbf{u} + \mathbf{v}) = \mathbf{T}\mathbf{u} + \mathbf{T}\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$,
- (2) $\mathbf{T}(\alpha\mathbf{u}) = \alpha\mathbf{T}\mathbf{u}$ for all $\alpha \in \mathbb{R}$ and $\mathbf{u} \in \mathcal{V}$.

We denote the set of all second-order tensors on $\mathcal{V}$ by the symbol $\mathcal{V}^2$. Analogous to the zero vector, we define a **zero tensor** $\mathbf{O}$ with the property $\mathbf{O}\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathcal{V}$, and we define an **identity tensor** $\mathbf{I}$ with the property $\mathbf{I}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathcal{V}$. Two second-order tensors $\mathbf{S}$ and $\mathbf{T}$ are said to be equal if and only if $\mathbf{S}\mathbf{v} = \mathbf{T}\mathbf{v}$ for all $\mathbf{v} \in \mathcal{V}$.

Examples:

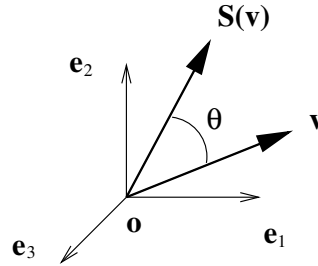
(1) **Projections.** The action of projecting a vector $\mathbf{v}$ onto a given unit vector $\mathbf{e}$ can be expressed in terms of a second-order tensor $\mathbf{P}$, namely

$$\mathbf{P}(\mathbf{v}) = (\mathbf{v} \cdot \mathbf{e})\mathbf{e}.$$



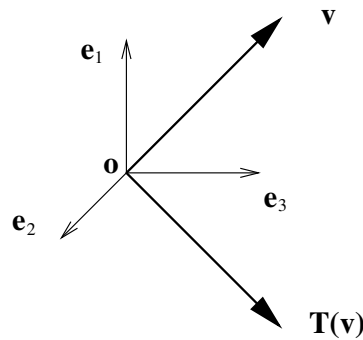
(2) **Rotations.** The action of rotating a vector $\mathbf{v}$ through a given angle θ about a fixed axis, say $\mathbf{e}_3$, can be expressed in terms of a second-order tensor $\mathbf{S}$. In particular, for each of the basis vectors $\mathbf{e}_i$ we have

$$\mathbf{S}(\mathbf{e}_1) = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2, \quad \mathbf{S}(\mathbf{e}_2) = \cos \theta \mathbf{e}_2 - \sin \theta \mathbf{e}_1, \quad \mathbf{S}(\mathbf{e}_3) = \mathbf{e}_3.$$



(3) **Reflections.** The action of reflecting a vector $\mathbf{v}$ with respect to a fixed plane, say with normal $\mathbf{e}_1$, can be expressed in terms of a second-order tensor $\mathbf{T}$, namely

$$\mathbf{T}(\mathbf{v}) = \mathbf{v} - 2(\mathbf{v} \cdot \mathbf{e}_1)\mathbf{e}_1.$$



□

1.3.2 Second-Order Tensor Algebra

If we let $\mathcal{V}^2$ be the set of second-order tensors, then $\mathcal{V}^2$ has the structure of a real vector space since

$$\mathbf{S} + \mathbf{T} \in \mathcal{V}^2 \quad \forall \mathbf{S}, \mathbf{T} \in \mathcal{V}^2 \quad \text{and} \quad \alpha \mathbf{T} \in \mathcal{V}^2 \quad \forall \alpha \in \mathbb{R}, \mathbf{T} \in \mathcal{V}^2.$$

We can also compose second-order tensors in the sense that

$$\mathbf{ST} \in \mathcal{V}^2 \quad \forall \mathbf{S}, \mathbf{T} \in \mathcal{V}^2.$$

In particular, we define the sum $\mathbf{S} + \mathbf{T}$ via the relation $(\mathbf{S} + \mathbf{T})\mathbf{v} = \mathbf{S}\mathbf{v} + \mathbf{T}\mathbf{v}$ for all $\mathbf{v} \in \mathcal{V}$, and we define the composition $\mathbf{ST}$ via the relation $(\mathbf{ST})\mathbf{v} = \mathbf{S}(\mathbf{T}\mathbf{v})$ for all $\mathbf{v} \in \mathcal{V}$. A similar definition holds for $\alpha\mathbf{T}$.

1.3.3 Representation in a Coordinate Frame

By the **components** of a second-order tensor $\mathbf{S}$ in a coordinate frame $\{\mathbf{e}_i\}$ we mean the nine numbers S_{ij} ($1 \leq i, j \leq 3$) defined by

$$S_{ij} = \mathbf{e}_i \cdot \mathbf{S}\mathbf{e}_j.$$

These components completely describe the transformation $\mathbf{S} : \mathcal{V} \rightarrow \mathcal{V}$. Given vectors $\mathbf{v} = v_i \mathbf{e}_i$ and $\mathbf{u} = u_j \mathbf{e}_j$ such that $\mathbf{v} = \mathbf{S}\mathbf{u}$ we have, by linearity

$$v_i = \mathbf{e}_i \cdot \mathbf{S}\mathbf{u} = (\mathbf{e}_i \cdot \mathbf{S}\mathbf{e}_j)u_j = S_{ij}u_j. \quad (1.6)$$

Thus the components of $\mathbf{S}$ are simply the coefficients in the linear relation between the components of $\mathbf{v}$ and $\mathbf{u}$.

Frequently, we arrange the nine components S_{ij} into a square matrix $[\mathbf{S}]$, in particular

$$[\mathbf{S}] = \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{pmatrix} \in \mathbb{R}^{3 \times 3},$$

where $\mathbb{R}^{3 \times 3}$ denotes the set of all 3×3 matrices of real numbers. We call $[\mathbf{S}]$ the matrix **representation** of $\mathbf{S}$ in the given coordinate frame. We could also arrange the components into a 9×1 matrix, that is, an element of $\mathbb{R}^9$. However, the square matrix representation will be more useful for our developments.

If we denote the element of $[\mathbf{S}]$ at the i th-row and j th-column by $[\mathbf{S}]_{ij}$, then we have $[\mathbf{S}]_{ij} = S_{ij}$. The **transpose** of the matrix $[\mathbf{S}]$ is denoted

by $[\mathbf{S}]^T$, and its elements are defined by the expression $[\mathbf{S}]_{ij}^T = [\mathbf{S}]_{ji}$. From the definitions of $[\mathbf{S}]$ and S_{ij} we deduce the convenient expression

$$[\mathbf{S}] = ([\mathbf{S}e_1], [\mathbf{S}e_2], [\mathbf{S}e_3]),$$

where $[\mathbf{S}e_j]$ is the 3×1 column matrix of components of $\mathbf{S}e_j$.

In view of (1.6) and the rules of matrix multiplication, the tensor equation $\mathbf{v} = \mathbf{S}\mathbf{u}$ can be written in various equivalent ways

$$\mathbf{v} = \mathbf{S}\mathbf{u} \quad \Leftrightarrow \quad v_i = S_{ij}u_j \quad \Leftrightarrow \quad [\mathbf{v}] = [\mathbf{S}][\mathbf{u}].$$

Thus the matrix representations of vectors and second-order tensors have the property

$$[\mathbf{S}\mathbf{u}] = [\mathbf{S}][\mathbf{u}].$$

Examples:

- (1) Let $\mathbf{S}$ and $\mathbf{T}$ be the rotation and reflection tensors introduced in the examples of Section 1.3.1. In the respective coordinate frame of each example, the matrix representation of $\mathbf{S}$ is

$$[\mathbf{S}] = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and the matrix representation of $\mathbf{T}$ is

$$[\mathbf{T}] = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (2) It is straightforward to show that, in any coordinate frame, the zero tensor $\mathbf{O}$ has components $O_{ij} = 0$ ($1 \leq i, j \leq 3$), and the identity tensor $\mathbf{I}$ has components $I_{ij} = \delta_{ij}$ ($1 \leq i, j \leq 3$).

□

1.3.4 Second-Order Dyadic Products, Bases

The **dyadic product** of two vectors $\mathbf{a}$ and $\mathbf{b}$ is the second-order tensor $\mathbf{a} \otimes \mathbf{b}$ defined by

$$(\mathbf{a} \otimes \mathbf{b})\mathbf{v} = (\mathbf{b} \cdot \mathbf{v})\mathbf{a}, \quad \forall \mathbf{v} \in \mathcal{V}.$$

In terms of components $[\mathbf{a} \otimes \mathbf{b}]_{ij}$, the above equation is equivalent to

$$[\mathbf{a} \otimes \mathbf{b}]_{ij} v_j = (b_j v_j) a_i, \quad \forall \mathbf{v} \in \mathcal{V},$$

which implies

$$[\mathbf{a} \otimes \mathbf{b}]_{ij} = a_i b_j.$$

To complete this last step simply take $\mathbf{v} = \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in turn. Throughout our developments we will move freely from similar expressions, which hold for all $\mathbf{v}$, to the statement with $\mathbf{v}$ removed. Notice that the matrix representation of $\mathbf{a} \otimes \mathbf{b}$ is

$$[\mathbf{a} \otimes \mathbf{b}] = \begin{pmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{pmatrix} = [\mathbf{a}][\mathbf{b}]^T.$$

Given any coordinate frame $\{\mathbf{e}_i\}$ the nine elementary dyadic products $\{\mathbf{e}_i \otimes \mathbf{e}_j\}$ form a **basis** for $\mathcal{V}^2$. In particular, each $\mathbf{S} \in \mathcal{V}^2$ can be uniquely represented as a linear combination

$$\mathbf{S} = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j, \quad (1.7)$$

where $S_{ij} = \mathbf{e}_i \cdot \mathbf{S} \mathbf{e}_j$. To see that the above representation is correct consider the expression $\mathbf{v} = \mathbf{S} \mathbf{u}$, where $\mathbf{v} = v_i \mathbf{e}_i$ and $\mathbf{u} = u_k \mathbf{e}_k$. Then

$$\begin{aligned} \mathbf{v} &= \mathbf{S} \mathbf{u} \\ &= S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j u_k \mathbf{e}_k \\ &= S_{ij} \delta_{jk} u_k \mathbf{e}_i \quad (\text{by dyadic property}) \\ &= S_{ij} u_j \mathbf{e}_i, \end{aligned}$$

which implies $v_i = S_{ij} u_j$. Thus (1.7) agrees with (1.6) and provides a unique representation of $\mathbf{S}$ in terms of its components S_{ij} .

1.3.5 Second-Order Tensor Algebra in Components

Let $\mathbf{S} = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ and $\mathbf{T} = T_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ be second-order tensors with matrix representations $[\mathbf{S}]$ and $[\mathbf{T}]$. Then from the definitions of $\mathbf{S} + \mathbf{T}$ and $\alpha \mathbf{T}$ it is straightforward to deduce

$$[\mathbf{S} + \mathbf{T}] = [\mathbf{S}] + [\mathbf{T}] \quad \text{and} \quad [\alpha \mathbf{T}] = \alpha [\mathbf{T}].$$

To establish a component expression for composition consider any two

dyadic products $\mathbf{a} \otimes \mathbf{b}$ and $\mathbf{c} \otimes \mathbf{d}$. Then for any arbitrary vector $\mathbf{v}$ we have

$$\begin{aligned} ((\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d}))\mathbf{v} &= (\mathbf{a} \otimes \mathbf{b})((\mathbf{c} \otimes \mathbf{d})\mathbf{v}) \\ &= (\mathbf{a} \otimes \mathbf{b})\mathbf{c}(\mathbf{d} \cdot \mathbf{v}) \\ &= \mathbf{a}(\mathbf{b} \cdot \mathbf{c})(\mathbf{d} \cdot \mathbf{v}) \\ &= (\mathbf{b} \cdot \mathbf{c})(\mathbf{a} \otimes \mathbf{d})\mathbf{v}, \end{aligned}$$

which, by the arbitrariness of $\mathbf{v}$, implies

$$(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d}) = (\mathbf{b} \cdot \mathbf{c})\mathbf{a} \otimes \mathbf{d}.$$

With this result we can express tensor composition in terms of components as follows. Let $\mathbf{S}$ and $\mathbf{T}$ be as above and let $\mathbf{U} = \mathbf{ST}$. Then

$$\begin{aligned} \mathbf{U} &= (S_{ij}\mathbf{e}_i \otimes \mathbf{e}_j)(T_{kl}\mathbf{e}_k \otimes \mathbf{e}_l) \\ &= S_{ij}T_{kl}\delta_{jk}\mathbf{e}_i \otimes \mathbf{e}_l \\ &= S_{ij}T_{jl}\mathbf{e}_i \otimes \mathbf{e}_l, \end{aligned}$$

which implies that $\mathbf{U}$ has components $U_{il} = S_{ij}T_{jl}$. Notice that this corresponds to the standard notion of matrix multiplication for the representations of $\mathbf{S}$ and $\mathbf{T}$, that is

$$[\mathbf{U}] = [\mathbf{ST}] = [\mathbf{S}][\mathbf{T}].$$

1.3.6 Special Classes of Tensors

To any tensor $\mathbf{S} \in \mathcal{V}^2$ we associate a **transpose** $\mathbf{S}^T \in \mathcal{V}^2$, which is the unique tensor with the property

$$\mathbf{S}\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{S}^T\mathbf{v} \quad \forall \mathbf{u}, \mathbf{v} \in \mathcal{V}.$$

We say $\mathbf{S}$ is **symmetric** if $\mathbf{S}^T = \mathbf{S}$ and **skew-symmetric** if $\mathbf{S}^T = -\mathbf{S}$.

A tensor $\mathbf{S} \in \mathcal{V}^2$ is said to be **positive-definite** if it satisfies

$$\mathbf{v} \cdot \mathbf{S}\mathbf{v} > 0 \quad \forall \mathbf{v} \neq \mathbf{0},$$

and is said to be **invertible** if there exists an **inverse** $\mathbf{S}^{-1} \in \mathcal{V}^2$ such that

$$\mathbf{S}\mathbf{S}^{-1} = \mathbf{S}^{-1}\mathbf{S} = \mathbf{I}.$$

The operations of inverse and transpose commute, that is, $(\mathbf{S}^{-1})^T = (\mathbf{S}^T)^{-1}$, and we denote the resulting tensor by $\mathbf{S}^{-T}$.

A tensor $\mathbf{Q} \in \mathcal{V}^2$ is said to be **orthogonal** if $\mathbf{Q}^T = \mathbf{Q}^{-1}$, that is

$$\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}.$$

An orthogonal tensor is called a **rotation** if, given any right-handed orthonormal frame $\{\mathbf{e}_i\}$, the set of vectors $\{\mathbf{Q}\mathbf{e}_i\}$ is also a right-handed orthonormal frame. Later we will see that an orthogonal tensor is a rotation when it has positive determinant.

In any given coordinate frame the above ideas are just the standard notions of the transpose of a matrix, a symmetric matrix, a skew-symmetric matrix, a positive-definite matrix, the inverse of a matrix, an orthogonal matrix and a rotation matrix, respectively. Moreover, we have (see Exercise 11)

$$[\mathbf{S}^T] = [\mathbf{S}]^T, \quad [\mathbf{S}^{-1}] = [\mathbf{S}]^{-1}, \quad [\mathbf{S}^{-T}] = [\mathbf{S}]^{-T}.$$

The proof of the following result is straightforward.

Result 1.2 Symmetric-Skew Decomposition. *Every second-order tensor can be uniquely written as*

$$\mathbf{S} = \mathbf{E} + \mathbf{W},$$

where $\mathbf{E}$ is a symmetric tensor and $\mathbf{W}$ is a skew-symmetric tensor. Specifically, we have

$$\mathbf{E} = \frac{1}{2}(\mathbf{S} + \mathbf{S}^T) \quad \text{and} \quad \mathbf{W} = \frac{1}{2}(\mathbf{S} - \mathbf{S}^T).$$

□

The above result motivates the two mappings $\text{sym} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ and $\text{skew} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ defined by

$$\text{sym}(\mathbf{S}) = \frac{1}{2}(\mathbf{S} + \mathbf{S}^T) \quad \text{and} \quad \text{skew}(\mathbf{S}) = \frac{1}{2}(\mathbf{S} - \mathbf{S}^T).$$

It is straightforward to show that, in any coordinate frame, a symmetric tensor $\mathbf{E}$ satisfies $E_{ij} = E_{ji}$, and a skew-symmetric tensor $\mathbf{W}$ satisfies $W_{ij} = -W_{ji}$.

Result 1.3 Skew Tensor as Vector Product. *Given any skew-symmetric tensor $\mathbf{W} \in \mathcal{V}^2$ there is a unique vector $\mathbf{w} \in \mathcal{V}$ such that*

$$\mathbf{W}\mathbf{v} = \mathbf{w} \times \mathbf{v}, \quad \forall \mathbf{v} \in \mathcal{V}.$$

We write $\mathbf{w} = \text{vec}(\mathbf{W})$ and call it the **axial vector** associated with $\mathbf{W}$. In any frame we have

$$w_j = \frac{1}{2} \epsilon_{njm} W_{nm}.$$

Conversely, given any vector $\mathbf{w} \in \mathcal{V}$ there is a unique skew-symmetric tensor $\mathbf{W} \in \mathcal{V}^2$ such that

$$\mathbf{w} \times \mathbf{v} = \mathbf{W}\mathbf{v}, \quad \forall \mathbf{v} \in \mathcal{V}.$$

We write $\mathbf{W} = \text{ten}(\mathbf{w})$ and call it the **axial tensor** associated with $\mathbf{w}$. In any frame we have

$$W_{ik} = \epsilon_{ijk} w_j.$$

The functions $\mathbf{W} = \text{ten}(\mathbf{w})$ and $\mathbf{w} = \text{vec}(\mathbf{W})$ are inverses. □

Proof To establish the first result let $\mathbf{W}$ be given. Then we seek $\mathbf{w}$ such that $\mathbf{W}\mathbf{v} = \mathbf{w} \times \mathbf{v}$ for all $\mathbf{v}$. In components this equation reads

$$W_{ik} v_k = \epsilon_{ijk} w_j v_k,$$

and by the arbitrariness of v_k we deduce

$$W_{ik} = \epsilon_{ijk} w_j. \tag{1.8}$$

We next show that there exists a vector w_j which satisfies this equation, and then show that the vector is unique. To establish existence let

$$w_j = \frac{1}{2} \epsilon_{njm} W_{nm}. \tag{1.9}$$

Substituting this expression into the right-hand side of (1.8), and using Result 1.1 and the fact that $W_{ki} = -W_{ik}$, we obtain

$$\epsilon_{ijk} w_j = \frac{1}{2} \epsilon_{ijk} \epsilon_{njm} W_{nm} = \frac{1}{2} (\delta_{in} \delta_{km} - \delta_{im} \delta_{kn}) W_{nm} = W_{ik}.$$

Thus the vector w_j defined in (1.9) satisfies (1.8). To establish uniqueness let w_j and u_j be any two solutions. Then from (1.8) we deduce $0 = \epsilon_{ijk} (w_j - u_j)$ for $i, k = 1, 2, 3$. Considering this equation with $i = 1$ and $k = 3$ we find $w_2 = u_2$, and by similar considerations we find $w_1 = u_1$ and $w_3 = u_3$. Thus the solution in (1.9) is unique.

To establish the second result let $\mathbf{w}$ be given. Then we seek a skew-symmetric tensor $\mathbf{W}$ such that $\mathbf{w} \times \mathbf{v} = \mathbf{W}\mathbf{v}$ for all $\mathbf{v}$. In components this equation reads

$$\epsilon_{ijk} w_j v_k = W_{ik} v_k,$$

and by the arbitrariness of v_k we deduce

$$W_{ik} = \epsilon_{ijk} w_j. \tag{1.10}$$

This equation provides an explicit expression for the tensor W_{ik} , which

is skew-symmetric by properties of the permutation symbol. Uniqueness follows immediately since W_{ik} is explicitly determined by w_j .

To establish the final result let $\mathbf{w} = \text{vec}(\mathbf{W})$ be the map defined by (1.9) and let $\mathbf{W} = \text{ten}(\mathbf{w})$ be the map defined by (1.10). Then for any arbitrary skew-symmetric tensor $\mathbf{W}$ we have, using Result 1.1

$$\begin{aligned} [\text{ten}(\text{vec}(\mathbf{W}))]_{ik} &= [\text{ten}(\mathbf{w})]_{ik} \\ &= \epsilon_{ijk} w_j \\ &= \frac{1}{2} \epsilon_{ijk} \epsilon_{njm} W_{nm} \\ &= \frac{1}{2} (\delta_{in} \delta_{km} - \delta_{im} \delta_{kn}) W_{nm} \\ &= W_{ik}. \end{aligned}$$

Thus for any skew-symmetric tensor $\mathbf{W}$ we have $\text{ten}(\text{vec}(\mathbf{W})) = \mathbf{W}$. Similarly, for any vector $\mathbf{w}$ we find $\text{vec}(\text{ten}(\mathbf{w})) = \mathbf{w}$. Thus the functions $\mathbf{W} = \text{ten}(\mathbf{w})$ and $\mathbf{w} = \text{vec}(\mathbf{W})$ are inverses. $\square$

1.3.7 Change of Basis

In any given coordinate frame the representation of a vector $\mathbf{v} \in \mathcal{V}$ is a triplet of components $[\mathbf{v}] \in \mathbb{R}^3$, and the representation of a second-order tensor $\mathbf{S} \in \mathcal{V}^2$ is a matrix of components $[\mathbf{S}] \in \mathbb{R}^{3 \times 3}$. In this section we discuss how the representations $[\mathbf{v}]$ and $[\mathbf{S}]$ change when the coordinate frame is changed.

1.3.7.1 Change of Basis Tensors

Let $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$ be two coordinate frames for $\mathbb{E}^3$ as shown in Figure 1.4. By the **change of basis** tensor from $\{\mathbf{e}_i\}$ to $\{\mathbf{e}'_i\}$ we mean the tensor $\mathbf{A}$ defined by

$$\mathbf{A} = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j \quad \text{where} \quad A_{ij} = \mathbf{e}_i \cdot \mathbf{e}'_j. \quad (1.11)$$

We could also define a change of basis tensor $\mathbf{B}$ from $\{\mathbf{e}'_i\}$ to $\{\mathbf{e}_i\}$ by $\mathbf{B} = B_{ij} \mathbf{e}'_i \otimes \mathbf{e}'_j$ where $B_{ij} = \mathbf{e}'_i \cdot \mathbf{e}_j$. All that we say for $\mathbf{A}$ will also apply to $\mathbf{B}$. However, for convenience, we work only with $\mathbf{A}$.

Using the components of $\mathbf{A}$ we can express the basis vectors of one frame in terms of the other. For example, a basis vector $\mathbf{e}'_j$ may be expressed in the frame $\{\mathbf{e}_i\}$ as

$$\mathbf{e}'_j = (\mathbf{e}_1 \cdot \mathbf{e}'_j) \mathbf{e}_1 + (\mathbf{e}_2 \cdot \mathbf{e}'_j) \mathbf{e}_2 + (\mathbf{e}_3 \cdot \mathbf{e}'_j) \mathbf{e}_3 = (e_i \cdot e'_j) \mathbf{e}_i,$$

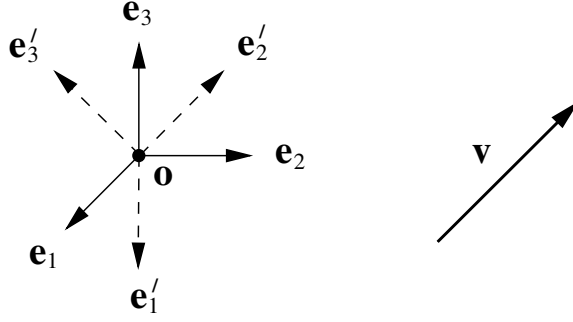


Fig. 1.4 Two different coordinate frames for describing vectors in space.

and by (1.11) this may be written as

$$\mathbf{e}'_j = A_{ij} \mathbf{e}_i. \quad (1.12)$$

Similarly, a basis vector $\mathbf{e}_i$ may be expressed in the frame $\{\mathbf{e}'_k\}$ as

$$\mathbf{e}_i = (\mathbf{e}_i \cdot \mathbf{e}'_k) \mathbf{e}'_k = A_{ik} \mathbf{e}'_k. \quad (1.13)$$

The above relations imply the following:

Result 1.4 Orthogonality of Basis Change Tensor. *The components A_{ij} of the change of basis tensor $\mathbf{A}$ have the property that*

$$A_{ij} A_{ik} = \delta_{jk} \quad \text{and} \quad A_{ik} A_{lk} = \delta_{il},$$

or in matrix notation

$$[\mathbf{A}]^T [\mathbf{A}] = [\mathbf{I}] \quad \text{and} \quad [\mathbf{A}] [\mathbf{A}]^T = [\mathbf{I}].$$

In particular, the change of basis tensor $\mathbf{A}$ is orthogonal. □

Proof Substituting (1.13) into (1.12) leads to

$$\mathbf{e}'_j = A_{ij} A_{ik} \mathbf{e}'_k,$$

which implies $A_{ij} A_{ik} = \delta_{jk}$. Similarly, substituting (1.12) into (1.13) yields $A_{ik} A_{lk} = \delta_{il}$. □

1.3.7.2 Change of Representation

Consider an arbitrary vector $\mathbf{v} \in \mathcal{V}$ as illustrated in Figure 1.4. Let $[\mathbf{v}] = (v_1, v_2, v_3)^T$ be its representation in $\{\mathbf{e}_i\}$, let $[\mathbf{v}]' = (v'_1, v'_2, v'_3)^T$ be its representation in $\{\mathbf{e}'_i\}$, and let $\mathbf{A}$ be the change of basis tensor from $\{\mathbf{e}_i\}$ to $\{\mathbf{e}'_i\}$. Because v_i and v'_i are components of the same physical

vector they are related through A_{ij} . To see this, notice that by definition of components we have $\mathbf{v} = v_i \mathbf{e}_i = v'_j \mathbf{e}'_j$, and by definition of A_{ij} we have

$$\mathbf{v} = v_i \mathbf{e}_i = v'_j \mathbf{e}'_j = v'_j A_{ij} \mathbf{e}_i,$$

which implies $v_i = A_{ij} v'_j$. Similarly, $v'_k = A_{ik} v_i$. In matrix notation we write this as

$$[\mathbf{v}] = [\mathbf{A}][\mathbf{v}]' \quad \text{and} \quad [\mathbf{v}]' = [\mathbf{A}]^T [\mathbf{v}].$$

For second-order tensors we have

$$[\mathbf{S}] = [\mathbf{A}][\mathbf{S}'][\mathbf{A}]^T \quad \text{and} \quad [\mathbf{S}]' = [\mathbf{A}]^T [\mathbf{S}][\mathbf{A}]. \quad (1.14)$$

The derivation of these results is straightforward (see Exercise 13).

1.3.8 Traces, Determinants and Exponentials

The **trace** function on second-order tensors is a linear mapping $\text{tr} : \mathcal{V}^2 \rightarrow \mathbb{R}$ defined by

$$\text{tr } \mathbf{S} = \text{tr}[\mathbf{S}] = [\mathbf{S}]_{ii},$$

where $[\mathbf{S}]$ denotes a matrix representation in any coordinate frame. Thus the trace of a second-order tensor $\mathbf{S}$ is just the usual trace (sum of diagonal elements) of a matrix representation $[\mathbf{S}]$. The next result shows that this definition does not depend on the coordinate frame.

Result 1.5 Invariance of the Trace. *Let $\mathbf{S}$ be a second-order tensor with matrix representations $[\mathbf{S}]$ and $[\mathbf{S}]'$ in the frames $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$, respectively. Then*

$$\text{tr}[\mathbf{S}] = \text{tr}[\mathbf{S}]'.$$

Thus the numerical value of the trace is independent of the coordinate frame in which it is computed. □

Proof Let $\mathbf{A}$ be the change of basis tensor from $\{\mathbf{e}_i\}$ to $\{\mathbf{e}'_i\}$. Then by (1.14) we have $[\mathbf{S}] = [\mathbf{A}][\mathbf{S}'][\mathbf{A}]^T$ or $[\mathbf{S}]_{ij} = [\mathbf{A}]_{ik}[\mathbf{S}']_{kl}[\mathbf{A}]_{jl}$, and

$$\begin{aligned} \text{tr}[\mathbf{S}] &= [\mathbf{S}]_{ii} = [\mathbf{S}']_{kl}[\mathbf{A}]_{ik}[\mathbf{A}]_{il} \\ &= [\mathbf{S}']_{kl}\delta_{kl} && \text{(by Result 1.4)} \\ &= [\mathbf{S}']_{kk} \\ &= \text{tr}[\mathbf{S}']'. \end{aligned}$$

□

The **determinant** function on second-order tensors is a mapping $\det : \mathcal{V}^2 \rightarrow \mathbb{R}$ defined by

$$\det \mathbf{S} = \det[\mathbf{S}] = \epsilon_{ijk} [\mathbf{S}]_{i1} [\mathbf{S}]_{j2} [\mathbf{S}]_{k3},$$

where $[\mathbf{S}]$ denotes a matrix representation in any coordinate frame. Thus the determinant of a second-order tensor $\mathbf{S}$ is just the usual determinant of a matrix representation $[\mathbf{S}]$. The explicit expression in terms of the permutation symbol is a consequence of the relation in (1.4). The next result shows that this definition does not depend on the coordinate frame.

Result 1.6 Invariance of the Determinant. *Let $\mathbf{S}$ be a second-order tensor with matrix representations $[\mathbf{S}]$ and $[\mathbf{S}']$ in the frames $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$, respectively. Then*

$$\det[\mathbf{S}] = \det[\mathbf{S}'].$$

Thus the numerical value of the determinant is independent of the coordinate frame in which it is computed. □

Proof See Exercise 20. □

The quantity $|\det \mathbf{S}|$ has the geometric interpretation of being the volume of the parallelepiped defined by the three vectors $\mathbf{S}\mathbf{e}_1$, $\mathbf{S}\mathbf{e}_2$ and $\mathbf{S}\mathbf{e}_3$. The determinant also arises in the classification of certain types of tensors. For example, a tensor $\mathbf{S}$ is invertible if and only if $\det \mathbf{S} \neq 0$. Moreover, because $\det(\mathbf{AB}) = (\det \mathbf{A})(\det \mathbf{B})$, every orthogonal tensor $\mathbf{Q}$ has the property that $|\det \mathbf{Q}| = 1$, and rotation tensors are those orthogonal tensors which satisfy $\det \mathbf{Q} = 1$ (see Exercise 19).

The **exponential** function on second-order tensors is a mapping $\exp : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ defined by

$$\exp(\mathbf{S}) = \sum_{j=0}^{\infty} \frac{1}{j!} \mathbf{S}^j = \mathbf{I} + \mathbf{S} + \frac{1}{2} \mathbf{S}^2 + \cdots.$$

It can be shown that this series converges for any $\mathbf{S} \in \mathcal{V}^2$ and so the exponential function is well-defined. This function arises in the theory of linear, constant coefficient, ordinary differential equations. Our main interest in it is summarized in the following result, whose proof we omit for brevity.

Result 1.7 Exponential of a Skew Tensor. *Let $\mathbf{W} \in \mathcal{V}^2$ be an arbitrary skew-symmetric tensor. Then $\exp(\mathbf{W}) \in \mathcal{V}^2$ is a rotation in the sense that $(\exp(\mathbf{W}))^T \exp(\mathbf{W}) = \mathbf{I}$ and $\det(\exp(\mathbf{W})) = 1$. Moreover, $(\exp(\mathbf{W}))^T = \exp(\mathbf{W}^T)$.* $\square$

1.3.9 Eigenvalues, Eigenvectors and Principal Invariants

By an **eigenpair** for a second-order tensor $\mathbf{S}$ we mean a scalar λ and a unit vector $\mathbf{e}$ satisfying

$$\mathbf{S}\mathbf{e} = \lambda\mathbf{e}.$$

Any such λ is called an **eigenvalue** and any such $\mathbf{e}$ an **eigenvector** of $\mathbf{S}$. From linear algebra we recall that λ is an eigenvalue if and only if it is a root of the characteristic (cubic) polynomial

$$p(\lambda) = \det(\mathbf{S} - \lambda\mathbf{I}). \quad (1.15)$$

Moreover, to any eigenvalue λ we associate one or more independent eigenvectors $\mathbf{e}$ by solving (in components) the linear, homogeneous equation

$$(\mathbf{S} - \lambda\mathbf{I})\mathbf{e} = \mathbf{0}. \quad (1.16)$$

When $\mathbf{S}$ is symmetric it can be shown that all three roots (possibly repeated) of the characteristic polynomial in (1.15) are real. Consequently, all solutions of the homogeneous equation in (1.16) are also real. Thus every symmetric tensor has exactly three eigenvalues, these eigenvalues are real, and each distinct eigenvalue has one or more independent eigenvectors, which are also real. Throughout our developments we consider eigenvalues and eigenvectors only for symmetric tensors.

Result 1.8 Eigenvalues, Eigenvectors of Symmetric Tensors.

(1) *The eigenvalues of a symmetric, positive-definite tensor are strictly positive.* (2) *Any two eigenvectors corresponding to distinct eigenvalues of a symmetric tensor are orthogonal.* $\square$

Proof

- (1) Let $\mathbf{S}$ be symmetric, positive-definite and let $(\lambda, \mathbf{e})$ be an eigenpair. Then, since $\mathbf{S}\mathbf{e} = \lambda\mathbf{e}$ and $\mathbf{e}$ is a unit vector, we have

$$0 < \mathbf{e} \cdot \mathbf{S}\mathbf{e} = \lambda\mathbf{e} \cdot \mathbf{e} = \lambda.$$

- (2) Let $\mathbf{S}$ be symmetric and let $(\lambda, \mathbf{e})$ and $(\omega, \mathbf{d})$ be two eigenpairs such that $\lambda \neq \omega$. Then, since $\mathbf{S}\mathbf{e} = \lambda\mathbf{e}$ and $\mathbf{S}\mathbf{d} = \omega\mathbf{d}$, we have

$$\lambda \mathbf{e} \cdot \mathbf{d} = \mathbf{S}\mathbf{e} \cdot \mathbf{d} = \mathbf{e} \cdot \mathbf{S}^T \mathbf{d} = \mathbf{e} \cdot \mathbf{S}\mathbf{d} = \omega \mathbf{e} \cdot \mathbf{d},$$

which implies $(\lambda - \omega)\mathbf{e} \cdot \mathbf{d} = 0$. Since $\lambda \neq \omega$ we must have $\mathbf{e} \cdot \mathbf{d} = 0$. □

We state, without proof, the following result from linear algebra.

Result 1.9 Spectral Decomposition Theorem. *Let $\mathbf{S}$ be a symmetric second-order tensor. Then there exists a right-handed, orthonormal basis $\{\mathbf{e}_i\}$ for $\mathcal{V}$ consisting of eigenvectors of $\mathbf{S}$. The corresponding eigenvalues λ_i are the same (up to ordering) for any such basis and form the full set of eigenvalues of $\mathbf{S}$. The dyadic representation of $\mathbf{S}$ in any such basis is*

$$\mathbf{S} = \sum_{i=1}^3 \lambda_i \mathbf{e}_i \otimes \mathbf{e}_i,$$

and the matrix representation is

$$[\mathbf{S}] = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}.$$

□

The **principal invariants** of a second-order tensor $\mathbf{S}$ are three scalars defined by

$$I_1(\mathbf{S}) = \text{tr } \mathbf{S}, \quad I_2(\mathbf{S}) = \frac{1}{2} [(\text{tr } \mathbf{S})^2 - \text{tr}(\mathbf{S}^2)], \quad I_3(\mathbf{S}) = \det \mathbf{S}.$$

For any $\alpha \in \mathbb{R}$ the principal invariants satisfy the relation

$$\det(\mathbf{S} - \alpha \mathbf{I}) = -\alpha^3 + I_1(\mathbf{S})\alpha^2 - I_2(\mathbf{S})\alpha + I_3(\mathbf{S}),$$

which can be verified by working in any coordinate frame. If $\mathbf{S}$ is symmetric, then the principal invariants can be written as

$$I_1(\mathbf{S}) = \lambda_1 + \lambda_2 + \lambda_3, \\ I_2(\mathbf{S}) = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1, \quad I_3(\mathbf{S}) = \lambda_1\lambda_2\lambda_3,$$

where λ_i are the eigenvalues of $\mathbf{S}$ (see Exercise 23). The quantities $I_i(\mathbf{S})$ are called invariants because their values are independent of any particular coordinate frame in which they are computed. The invariance

properties of I_1 and I_2 follow from Result 1.5, and the invariance property of I_3 follows from Result 1.6 (see Exercise 24). For brevity, we will sometimes use the symbol $\mathcal{I}_S$ to denote the triplet $(I_1(S), I_2(S), I_3(S))$.

Result 1.10 Cayley–Hamilton Theorem. *Any second-order tensor S satisfies the relation*

$$S^3 - I_1(S)S^2 + I_2(S)S - I_3(S)I = O.$$

□

Proof See Exercise 25.

□

1.3.10 Special Decompositions

Result 1.11 Tensor Square Root. *Let C be a symmetric, positive-definite tensor. Then there is a unique symmetric, positive-definite tensor U such that $U^2 = C$. In particular*

$$U = \sum_{i=1}^3 \sqrt{\lambda_i} \mathbf{e}_i \otimes \mathbf{e}_i,$$

where $\{\mathbf{e}_i\}$ is any frame consisting of eigenvectors of C and $\lambda_i > 0$ are the corresponding eigenvalues of C . We usually write $U = \sqrt{C}$. □

Proof See Exercise 26.

□

Result 1.12 Polar Decomposition Theorem. *Let F be a second-order tensor with positive determinant. Then there exist **right** and **left polar decompositions** of F taking the form, respectively*

$$F = RU = VR,$$

where $U = \sqrt{F^T F}$ and $V = \sqrt{F F^T}$ are symmetric, positive-definite tensors and R is a rotation. □

Proof Since F has positive determinant it is invertible. Thus $F\mathbf{v} \neq \mathbf{0}$ for all $\mathbf{v} \neq \mathbf{0}$. Similarly, the transpose F^T is invertible and $F^T\mathbf{v} \neq \mathbf{0}$ for all $\mathbf{v} \neq \mathbf{0}$. From these observations we deduce

$$\begin{aligned} \mathbf{v} \cdot F^T F \mathbf{v} &= (F\mathbf{v}) \cdot (F\mathbf{v}) > 0 \quad \forall \mathbf{v} \neq \mathbf{0}, \\ \mathbf{v} \cdot F F^T \mathbf{v} &= (F^T \mathbf{v}) \cdot (F^T \mathbf{v}) > 0 \quad \forall \mathbf{v} \neq \mathbf{0}. \end{aligned}$$

Thus $\mathbf{F}^T \mathbf{F}$ and $\mathbf{F} \mathbf{F}^T$ are positive-definite and clearly symmetric, and we can define $\mathbf{U}$ and $\mathbf{V}$ using Result 1.11.

Next, consider the tensor $\mathbf{R} = \mathbf{F} \mathbf{U}^{-1}$. Since $\mathbf{R} \mathbf{U} = \mathbf{F}$ and $\det(\mathbf{R} \mathbf{U}) = (\det \mathbf{R})(\det \mathbf{U})$ by properties of determinants (see Exercise 16), we have

$$\det \mathbf{R} = (\det \mathbf{F}) / (\det \mathbf{U}).$$

From this we deduce $\det \mathbf{R} > 0$ since $\det \mathbf{F} > 0$ and $\det \mathbf{U} > 0$. Moreover

$$\mathbf{R}^T \mathbf{R} = \mathbf{U}^{-T} \mathbf{F}^T \mathbf{F} \mathbf{U}^{-1} = \mathbf{U}^{-1} \mathbf{U}^2 \mathbf{U}^{-1} = \mathbf{I}.$$

Thus $\mathbf{R}$ is a rotation as claimed and the right polar decomposition is established.

To establish the left polar decomposition let $\mathbf{Q} = \mathbf{V}^{-1} \mathbf{F}$. Then by arguments similar to those above we deduce that $\mathbf{Q}$ is also a rotation. Thus we have $\mathbf{F} = \mathbf{R} \mathbf{U} = \mathbf{V} \mathbf{Q}$. To show that $\mathbf{Q} = \mathbf{R}$ let $\mathbf{S}$ be the symmetric, positive-definite tensor given by $\mathbf{S} = \mathbf{Q}^T \mathbf{V} \mathbf{Q}$. Then $\mathbf{R} \mathbf{U} = \mathbf{Q} \mathbf{S}$ and $\mathbf{R} = \mathbf{Q} \mathbf{S} \mathbf{U}^{-1}$. Since $\mathbf{R}$ is a rotation we have $\mathbf{R}^{-1} = \mathbf{R}^T$, which implies $\mathbf{S}^2 = \mathbf{U}^2 = \mathbf{F}^T \mathbf{F}$. From the uniqueness of the tensor square root we deduce $\mathbf{S} = \mathbf{U}$, which implies $\mathbf{R} = \mathbf{Q}$. Thus $\mathbf{F} = \mathbf{R} \mathbf{U} = \mathbf{V} \mathbf{R}$ as claimed. $\square$

1.3.11 Scalar Product for Second-Order Tensors

Analogous to the scalar product for vectors, we define a scalar (inner) product for second-order tensors by

$$\mathbf{S} : \mathbf{D} = \text{tr}(\mathbf{S}^T \mathbf{D}). \quad (1.17)$$

In any frame the dyadic basis $\{\mathbf{e}_i \otimes \mathbf{e}_j\}$ for $\mathcal{V}^2$ introduced in Section 1.3.4 is actually orthonormal in this scalar product (see Exercise 28).

Result 1.13 Tensor Scalar Product. *Let $\{\mathbf{e}_i\}$ be a coordinate frame and let $\mathbf{S} = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ and $\mathbf{D} = D_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ be any two second-order tensors. Then*

$$\mathbf{S} : \mathbf{D} = S_{ij} D_{ij}.$$

Furthermore, if $\mathbf{S}$ is symmetric, then

$$\mathbf{S} : \mathbf{D} = \mathbf{S} : \text{sym}(\mathbf{D}) = \text{sym}(\mathbf{S}) : \text{sym}(\mathbf{D}).$$

$\square$

Proof See Exercise 29. $\square$

1.4 Fourth-Order Tensors

In the previous section we defined a second-order tensor as a linear transformation between vectors. In describing the behavior of material bodies we will also need the concept of a *linear transformation between second-order tensors*. For example, the relationship between stress and strain in a material body is sometimes modeled by such a transformation. Linear transformations between second-order tensors lead to the notion of a fourth-order tensor as defined below. As we will see, a fourth-order tensor will be described by eighty-one components in any Cartesian coordinate frame for $\mathbb{E}^3$.

1.4.1 Definition

By a **fourth-order tensor** $\mathbf{C}$ on the vector space $\mathcal{V}$ we mean a mapping $\mathbf{C} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ which is *linear* in the sense that:

- (1) $\mathbf{C}(\mathbf{S} + \mathbf{T}) = \mathbf{C}\mathbf{S} + \mathbf{C}\mathbf{T}$ for all $\mathbf{S}, \mathbf{T} \in \mathcal{V}^2$,
- (2) $\mathbf{C}(\alpha\mathbf{T}) = \alpha\mathbf{C}\mathbf{T}$ for all $\alpha \in \mathbb{R}$ and $\mathbf{T} \in \mathcal{V}^2$.

We denote the set of all fourth-order tensors on $\mathcal{V}$ by the symbol $\mathcal{V}^4$. We define a fourth-order **zero tensor** $\mathbf{O}$ with the property $\mathbf{O}\mathbf{T} = \mathbf{O}$ for all $\mathbf{T} \in \mathcal{V}^2$, and we define a fourth-order **identity tensor** $\mathbf{I}$ with the property $\mathbf{I}\mathbf{T} = \mathbf{T}$ for all $\mathbf{T} \in \mathcal{V}^2$. Two fourth-order tensors $\mathbf{C}$ and $\mathbf{D}$ are said to be equal if and only if $\mathbf{C}\mathbf{T} = \mathbf{D}\mathbf{T}$ for all $\mathbf{T} \in \mathcal{V}^2$.

Example: For any fixed $\mathbf{A} \in \mathcal{V}^2$ the mapping $\mathbf{C} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ given by

$$\mathbf{C}(\mathbf{T}) = \mathbf{A}\mathbf{T}$$

defines a fourth-order tensor. To see this we need only establish the two properties listed above, but from the algebra of second-order tensors we immediately have

$$\begin{aligned} \mathbf{C}(\alpha\mathbf{S} + \beta\mathbf{T}) &= \mathbf{A}(\alpha\mathbf{S} + \beta\mathbf{T}) \\ &= \alpha\mathbf{A}\mathbf{S} + \beta\mathbf{A}\mathbf{T} = \alpha\mathbf{C}(\mathbf{S}) + \beta\mathbf{C}(\mathbf{T}), \end{aligned}$$

for any $\alpha, \beta \in \mathbb{R}$ and $\mathbf{S}, \mathbf{T} \in \mathcal{V}^2$. □

1.4.2 Fourth-Order Tensor Algebra

If we let $\mathcal{V}^4$ be the set of fourth-order tensors, then $\mathcal{V}^4$ has the structure of a real vector space since

$$\mathbf{C} + \mathbf{D} \in \mathcal{V}^4 \quad \forall \mathbf{C}, \mathbf{D} \in \mathcal{V}^4 \quad \text{and} \quad \alpha \mathbf{C} \in \mathcal{V}^4 \quad \forall \alpha \in \mathbb{R}, \mathbf{C} \in \mathcal{V}^4.$$

We can also compose fourth-order tensors in the sense that

$$\mathbf{CD} \in \mathcal{V}^4 \quad \forall \mathbf{C}, \mathbf{D} \in \mathcal{V}^4.$$

Similar to the case of second-order tensors, we define the sum $\mathbf{C} + \mathbf{D}$ via the relation $(\mathbf{C} + \mathbf{D})\mathbf{T} = \mathbf{CT} + \mathbf{DT}$ for all $\mathbf{T} \in \mathcal{V}^2$, and we define the composition $\mathbf{CD}$ via the relation $(\mathbf{CD})\mathbf{T} = \mathbf{C}(\mathbf{DT})$ for all $\mathbf{T} \in \mathcal{V}^2$. A similar definition holds for $\alpha\mathbf{C}$.

1.4.3 Representation in a Coordinate Frame

By the **components** of a fourth-order tensor $\mathbf{C}$ in a coordinate frame $\{\mathbf{e}_i\}$ we mean the eighty-one numbers C_{ijkl} ($1 \leq i, j, k, l \leq 3$) defined by

$$C_{ijkl} = \mathbf{e}_i \cdot \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l) \mathbf{e}_j.$$

That is, if we let $\mathbf{S}_{kl}$ be the second-order tensor given by $\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l)$, then

$$C_{ijkl} = \mathbf{e}_i \cdot \mathbf{S}_{kl} \mathbf{e}_j.$$

These components completely describe the transformation $\mathbf{C} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$. Given $\mathbf{U} = U_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ and $\mathbf{T} = T_{kl} \mathbf{e}_k \otimes \mathbf{e}_l$ related by $\mathbf{U} = \mathbf{C}(\mathbf{T})$, we have, by linearity

$$\begin{aligned} U_{ij} &= \mathbf{e}_i \cdot \mathbf{U} \mathbf{e}_j = \mathbf{e}_i \cdot \mathbf{C}(\mathbf{T}) \mathbf{e}_j \\ &= \mathbf{e}_i \cdot \mathbf{C}(T_{kl} \mathbf{e}_k \otimes \mathbf{e}_l) \mathbf{e}_j \\ &= \mathbf{e}_i \cdot \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l) \mathbf{e}_j T_{kl} \\ &= C_{ijkl} T_{kl}. \end{aligned} \tag{1.18}$$

Thus the components of $\mathbf{C}$ are simply the coefficients in the linear relation between the components of $\mathbf{U}$ and $\mathbf{T}$.

Example: Let $\mathbf{A} \in \mathcal{V}^2$ be given and consider the fourth-order tensor defined by $\mathbf{C}(\mathbf{T}) = \mathbf{AT}$. Then the components of $\mathbf{C}$ are

$$C_{ijkl} = \mathbf{e}_i \cdot \mathbf{A}(\mathbf{e}_k \otimes \mathbf{e}_l) \mathbf{e}_j = \mathbf{e}_i \cdot \mathbf{A} \mathbf{e}_k \delta_{lj} = A_{ik} \delta_{lj}.$$

□

An alternative way to compute the components C_{ijkl} is to use the scalar product for second-order tensors, namely

$$C_{ijkl} = (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l).$$

To see this, let $\mathbf{S}_{kl}$ be the second-order tensor given by $\mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l)$ and notice that

$$\begin{aligned} \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l) &= \mathbf{S}_{kl} \\ &= (\mathbf{e}_i \cdot \mathbf{S}_{kl} \mathbf{e}_j) \mathbf{e}_i \otimes \mathbf{e}_j \\ &= C_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j. \end{aligned}$$

Thus

$$\begin{aligned} (\mathbf{e}_m \otimes \mathbf{e}_n) : \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l) &= C_{ijkl} (\mathbf{e}_m \otimes \mathbf{e}_n) : (\mathbf{e}_i \otimes \mathbf{e}_j) \\ &= C_{ijkl} \delta_{mi} \delta_{nj} \\ &= C_{mnkl}, \end{aligned}$$

as required.

Just as for second-order tensors, the components of a fourth-order tensor may be arranged into various matrix representations. For example, if we arrange the nine components of a second-order tensor $\mathbf{T}$ into a 9×1 column matrix $[[\mathbf{T}]] \in \mathbb{R}^9$, then the eighty-one components of a fourth-order tensor $\mathbf{C}$ could naturally be arranged into a 9×9 matrix $[[\mathbf{C}]] \in \mathbb{R}^{9 \times 9}$. In this case the tensor equation $\mathbf{U} = \mathbf{C}\mathbf{T}$ would be equivalent to the matrix equation $[[\mathbf{U}]] = [[\mathbf{C}]] [[\mathbf{T}]]$. However, such matrix representations will not be exploited in our developments.

1.4.4 Fourth-Order Dyadic Products, Bases

The **dyadic product** of four vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ is the fourth-order tensor $\mathbf{a} \otimes \mathbf{b} \otimes \mathbf{c} \otimes \mathbf{d}$ defined by

$$(\mathbf{a} \otimes \mathbf{b} \otimes \mathbf{c} \otimes \mathbf{d})\mathbf{T} = (\mathbf{c} \cdot \mathbf{T}\mathbf{d})\mathbf{a} \otimes \mathbf{b}, \quad \forall \mathbf{T} \in \mathcal{V}^2.$$

Given any coordinate frame $\{\mathbf{e}_i\}$ the eighty-one elementary dyadic products $\{\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l\}$ form a **basis** for $\mathcal{V}^4$. In particular, each $\mathbf{C} \in \mathcal{V}^4$ can be uniquely represented as a linear combination

$$\mathbf{C} = C_{ijkl} \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l, \quad (1.19)$$

where $C_{ijkl} = \mathbf{e}_i \cdot \mathbf{C}(\mathbf{e}_k \otimes \mathbf{e}_l) \mathbf{e}_j$. To see that the above representation is correct consider the expression $\mathbf{U} = \mathbf{C}\mathbf{T}$, where $\mathbf{U} = U_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ and

$\mathbf{T} = T_{kl}\mathbf{e}_k \otimes \mathbf{e}_l$. Then

$$\begin{aligned} \mathbf{U} &= \mathbf{CT} \\ &= (\mathbf{C}_{ijkl}\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l)\mathbf{T} \\ &= \mathbf{C}_{ijkl}(\mathbf{e}_k \cdot \mathbf{T}\mathbf{e}_l)\mathbf{e}_i \otimes \mathbf{e}_j \\ &= \mathbf{C}_{ijkl}T_{kl}\mathbf{e}_i \otimes \mathbf{e}_j, \end{aligned}$$

which implies $U_{ij} = \mathbf{C}_{ijkl}T_{kl}$. Thus (1.19) agrees with (1.18) and provides a unique representation of $\mathbf{C}$ in terms of its components $\mathbf{C}_{ijkl}$.

1.4.5 Symmetry Properties

A fourth-order tensor $\mathbf{C} \in \mathcal{V}^4$ is said to be **symmetric** or possess **major symmetry** if

$$\mathbf{A} : \mathbf{C}(\mathbf{B}) = \mathbf{C}(\mathbf{A}) : \mathbf{B}, \quad \forall \mathbf{A}, \mathbf{B} \in \mathcal{V}^2.$$

It is said to possess a **right minor symmetry** if

$$\mathbf{A} : \mathbf{C}(\mathbf{B}) = \mathbf{A} : \mathbf{C}(\text{sym}(\mathbf{B})), \quad \forall \mathbf{A}, \mathbf{B} \in \mathcal{V}^2,$$

and a **left minor symmetry** if

$$\mathbf{A} : \mathbf{C}(\mathbf{B}) = \text{sym}(\mathbf{A}) : \mathbf{C}(\mathbf{B}), \quad \forall \mathbf{A}, \mathbf{B} \in \mathcal{V}^2.$$

The implications of the above conditions on the components of $\mathbf{C}$ are established in the following result.

Result 1.14 Fourth-order Tensor Symmetries. *If a fourth-order tensor $\mathbf{C}$ has major symmetry, then*

$$\mathbf{C}_{ijkl} = \mathbf{C}_{klij}.$$

If $\mathbf{C}$ has right and left minor symmetries, then

$$\mathbf{C}_{ijkl} = \mathbf{C}_{ijlk} \quad \text{and} \quad \mathbf{C}_{ijkl} = \mathbf{C}_{jikl},$$

respectively.

□

Proof See Exercise 33.

□

1.5 Isotropic Tensor Functions

A function $\mathbf{G} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is said to be **isotropic** if it satisfies

$$\mathbf{Q}\mathbf{G}(\mathbf{A})\mathbf{Q}^T = \mathbf{G}(\mathbf{Q}\mathbf{A}\mathbf{Q}^T),$$

for all $\mathbf{A} \in \mathcal{V}^2$ and all rotations $\mathbf{Q}$. Similarly, a function $g : \mathcal{V}^2 \rightarrow \mathbb{R}$ is said to be **isotropic** if it satisfies

$$g(\mathbf{A}) = g(\mathbf{Q}\mathbf{A}\mathbf{Q}^T),$$

for all $\mathbf{A} \in \mathcal{V}^2$ and all rotations $\mathbf{Q}$.

The condition of isotropy imposes severe restriction on functions. The next result, which we state without proof, shows that any isotropic function which maps the set of symmetric tensors into itself must have a particularly simple form. Recall the definition of, and notation for, principal invariants.

Result 1.15 Representation of Isotropic Tensor Functions. *Let $\mathbf{G} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ be an isotropic function which maps symmetric tensors to symmetric tensors, that is, $\mathbf{G}(\mathbf{A}) \in \mathcal{V}^2$ is symmetric if $\mathbf{A} \in \mathcal{V}^2$ is symmetric. Then there are functions $\alpha_0, \alpha_1, \alpha_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that*

$$\mathbf{G}(\mathbf{A}) = \alpha_0(\mathcal{I}_A)\mathbf{I} + \alpha_1(\mathcal{I}_A)\mathbf{A} + \alpha_2(\mathcal{I}_A)\mathbf{A}^2$$

for every symmetric $\mathbf{A} \in \mathcal{V}^2$. Consequently, by Result 1.10, there are functions $\beta_0, \beta_1, \beta_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$\mathbf{G}(\mathbf{A}) = \beta_0(\mathcal{I}_A)\mathbf{I} + \beta_1(\mathcal{I}_A)\mathbf{A} + \beta_2(\mathcal{I}_A)\mathbf{A}^{-1}$$

for every symmetric $\mathbf{A} \in \mathcal{V}^2$ which is invertible. □

The above result may be strengthened when, in addition to the properties mentioned above, $\mathbf{G}$ is linear and satisfies $\mathbf{G}(\mathbf{W}) = \mathbf{O}$ for every skew-symmetric $\mathbf{W}$. This case, which we now study, will be of particular importance in Chapters 6 and 7 when discussing linear models for stress in fluid and solid bodies.

Result 1.16 Isotropic Fourth-Order Tensors. *Let $\mathbf{C} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ be an isotropic fourth-order tensor with the properties:*

- (1) $\mathbf{C}(\mathbf{A}) \in \mathcal{V}^2$ is symmetric for every symmetric $\mathbf{A} \in \mathcal{V}^2$,
- (2) $\mathbf{C}(\mathbf{W}) = \mathbf{O}$ for every skew-symmetric $\mathbf{W} \in \mathcal{V}^2$.

Then there are scalars μ and λ such that

$$\mathbf{C}(\mathbf{A}) = \lambda(\text{tr } \mathbf{A})\mathbf{I} + 2\mu \text{sym}(\mathbf{A}), \quad \forall \mathbf{A} \in \mathcal{V}^2.$$

□

Proof Since $\mathbf{C} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is isotropic and maps symmetric tensors to symmetric tensors it follows from Result 1.15 that there are functions $\alpha_0, \alpha_1, \alpha_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$\mathbf{C}(\mathbf{H}) = \alpha_0(\mathcal{I}_H)\mathbf{I} + \alpha_1(\mathcal{I}_H)\mathbf{H} + \alpha_2(\mathcal{I}_H)\mathbf{H}^2$$

for every symmetric $\mathbf{H} \in \mathcal{V}^2$, where

$$\mathcal{I}_H = \left(\text{tr } \mathbf{H}, \frac{1}{2}[(\text{tr } \mathbf{H})^2 - \text{tr}(\mathbf{H}^2)], \det \mathbf{H} \right).$$

Since $\mathbf{C}(\mathbf{H})$ is linear in $\mathbf{H}$ the only possibilities are

$$\alpha_0(\mathcal{I}_H) = c_0 \text{tr } \mathbf{H} + c_1,$$

$$\alpha_1(\mathcal{I}_H) = c_2,$$

$$\alpha_2(\mathcal{I}_H) = 0,$$

where c_0, c_1 and c_2 are scalar constants. Since $\mathbf{C}(\mathbf{O}) = \mathbf{O}$ it follows that $c_1 = 0$. Thus, upon setting $\lambda = c_0$ and $\mu = c_2/2$, we have

$$\mathbf{C}(\mathbf{H}) = \lambda(\text{tr } \mathbf{H})\mathbf{I} + 2\mu\mathbf{H}$$

for every symmetric $\mathbf{H}$. This together with the fact that $\mathbf{C}(\mathbf{W}) = \mathbf{O}$ for every skew-symmetric $\mathbf{W}$ leads to the expression

$$\mathbf{C}(\mathbf{H}) = \lambda(\text{tr } \mathbf{H})\mathbf{I} + 2\mu \text{sym}(\mathbf{H}), \quad \forall \mathbf{H} \in \mathcal{V}^2.$$

□

Bibliographic Notes

Classic references for the material presented here are the texts by Bowen and Wang (1976) and Halmos (1974). A more modern treatment with applications to continuum mechanics is given in Knowles (1998). For simplicity of exposition we have chosen to work only in orthonormal coordinate frames. However, non-orthonormal coordinate frames arise naturally in many applications. A concise treatment of tensors in non-orthonormal frames can be found in Simmonds (1994), and a similar treatment from a more engineering point of view can be found in Malvern (1969).

The idea of a tensor can be defined in various different, but consistent ways. For example, many authors define an n th-order tensor on a vector space $\mathcal{V}$ to be a multilinear function from $\mathcal{V} \times \cdots \times \mathcal{V}$ (n copies) into $\mathbb{R}$. In this respect, a vector $\mathbf{v}$ is a first-order tensor because it can naturally (via the scalar product) be identified with a linear function $f : \mathcal{V} \rightarrow \mathbb{R}$, namely $f(\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}$. Similarly, a second-order tensor $\mathbf{S}$ as defined here can naturally be identified with a bilinear function $f : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$, namely $f(\mathbf{w}, \mathbf{u}) = \mathbf{w} \cdot \mathbf{S}\mathbf{u}$, and so on. The definitions of second- and fourth-order tensors given in this chapter as linear transformations are consistent with this more general definition. One reason for using the definition in terms of linear transformations is that it leads naturally to the idea of tensor components and matrix representations.

Vectors and tensors are of fundamental importance in mechanics and will be used throughout the remainder of our developments. In particular, vectors and tensors give us a way to state physical laws without reference to any particular coordinate frame. Various such laws for continuum bodies will be discussed in Chapter 5. A modern and more thorough discussion of tensors can be found in Bourne and Kendall (1992). For applications of tensors in differential geometry and analytical mechanics see Dodson and Poston (1991) and Abraham and Marsden (1978). For a general discussion of the history of vectors and tensors see Crowe (1967).

The theory of rotations, the exponential function and general isotropic functions are of special interest across various disciplines. A wealth of information on these subjects can be found, respectively, in Hughes (1986), Hirsch and Smale (1974) and Gurtin (1981). Proofs of our Results 1.7 and 1.15 can be found in Gurtin (1981), and proofs of our Results 1.9 and 1.10 can be found in both Bowen and Wang (1976) and Halmos (1974).

Exercises

1.1 Given the vectors $\mathbf{a} = 1\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = 1\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{c} = -2\mathbf{i} - 1\mathbf{j} + 0\mathbf{k}$, calculate:

- (a) $\mathbf{a} \cdot \mathbf{b}$,
- (b) $\mathbf{a} \times \mathbf{b}$,
- (c) $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$,
- (d) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$,
- (e) $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$.

- 1.2 Given a plane Π with normal $\mathbf{n} = 1\mathbf{i} - 2\mathbf{j} + 1\mathbf{k}$ and the vector $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, calculate:
- (a) the projection of $\mathbf{v}$ onto Π ,
 - (b) the reflection of $\mathbf{v}$ with respect to Π .

- 1.3 Calculate $\delta_{ij}\delta_{ij}$ using the rules of index notation and the definition of the Kronecker delta.

- 1.4 Suppose a vector $\mathbf{v}$ satisfies the linear equation

$$\alpha\mathbf{v} + \mathbf{v} \times \mathbf{a} = \mathbf{b},$$

where $\alpha \neq 0$ is a given scalar, and $\mathbf{a}$ and $\mathbf{b}$ are given vectors. Use the dot and cross product operations to solve the above equation for $\mathbf{v}$. In particular, show that the unique solution is given by

$$\mathbf{v} = \frac{\alpha^2\mathbf{b} - \alpha(\mathbf{b} \times \mathbf{a}) + (\mathbf{b} \cdot \mathbf{a})\mathbf{a}}{\alpha(\alpha^2 + |\mathbf{a}|^2)}.$$

- 1.5 Let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be vectors. Find scalars λ and μ such that

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \lambda\mathbf{b} - \mu\mathbf{a}.$$

- 1.6 Let $\mathbf{v}$ be an arbitrary vector and let $\mathbf{n}$ be an arbitrary unit vector. Show that:

- (a) $\mathbf{v} = (\mathbf{v} \cdot \mathbf{n})\mathbf{n} - (\mathbf{v} \times \mathbf{n}) \times \mathbf{n}$,
- (b) $\mathbf{v} = (\mathbf{v} \cdot \mathbf{n})\mathbf{n} + (\mathbf{v} \otimes \mathbf{n})\mathbf{n} - (\mathbf{n} \otimes \mathbf{v})\mathbf{n}$.

Remark: The identity in (a) shows that $\mathbf{v}$ can always be decomposed into parts parallel and perpendicular to $\mathbf{n}$.

- 1.7 Given two vectors $\mathbf{a}$ and $\mathbf{b}$, and a second-order tensor $\mathbf{S}$, prove:

- (a) $\mathbf{S}(\mathbf{a} \otimes \mathbf{b}) = (\mathbf{S}\mathbf{a}) \otimes \mathbf{b}$,
- (b) $(\mathbf{a} \otimes \mathbf{b})\mathbf{S} = \mathbf{a} \otimes (\mathbf{S}^T\mathbf{b})$,
- (c) $(\mathbf{a} \otimes \mathbf{b})^T = (\mathbf{b} \otimes \mathbf{a})$.

Hint: Recall that two second-order tensors $\mathbf{A}$ and $\mathbf{B}$ are equal if and only if $\mathbf{A}\mathbf{v} = \mathbf{B}\mathbf{v}$ for all $\mathbf{v} \in \mathcal{V}$.

- 1.8 Consider any three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ which are linearly independent, that is, $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} \neq 0$. Show that:

- (a) $\mathbf{a} \times \mathbf{b}$, $\mathbf{b} \times \mathbf{c}$ and $\mathbf{c} \times \mathbf{a}$ are also linearly independent,
- (b) $(\mathbf{a} \times \mathbf{b}) \otimes \mathbf{c} + (\mathbf{b} \times \mathbf{c}) \otimes \mathbf{a} + (\mathbf{c} \times \mathbf{a}) \otimes \mathbf{b} = (\mathbf{a} \times \mathbf{b} \cdot \mathbf{c})\mathbf{I}$.

- 1.9 A second-order tensor $\mathbf{P}$ is a **perpendicular projection** if $\mathbf{P}$ is symmetric and $\mathbf{P}^2 = \mathbf{P}$. Given two arbitrary unit vectors $\mathbf{n} \neq \mathbf{m}$, determine which of the following are perpendicular projections:

- (a) $\mathbf{P} = \mathbf{I}$,
- (b) $\mathbf{P} = \mathbf{n} \otimes \mathbf{m}$,
- (c) $\mathbf{P} = \mathbf{n} \otimes \mathbf{n}$,
- (d) $\mathbf{P} = \mathbf{I} - \mathbf{n} \otimes \mathbf{n}$,
- (e) $\mathbf{P} = \mathbf{n} \otimes \mathbf{m} + \mathbf{m} \otimes \mathbf{n}$.

- 1.10 Let $\mathbf{Q}$ be a second-order tensor and let $\mathbf{I}$ be the identity tensor. Show that $\mathbf{Q}$ is orthogonal if $\mathbf{H} = \mathbf{Q} - \mathbf{I}$ satisfies

$$\mathbf{H} + \mathbf{H}^T + \mathbf{H}\mathbf{H}^T = \mathbf{O}.$$

- 1.11 Show that the transpose of a second-order tensor $\mathbf{S}$ is uniquely defined and that $[\mathbf{S}^T] = [\mathbf{S}]^T$.

- 1.12 Prove that a second-order tensor $\mathbf{S}$ cannot be both positive-definite and skew-symmetric.

- 1.13 Let $\mathbf{A}$ denote the change of basis tensor from a frame $\{\mathbf{e}_i\}$ to a frame $\{\mathbf{e}'_i\}$ with representation $[\mathbf{A}]$ in $\{\mathbf{e}_i\}$. Let $\mathbf{S}$ be a second-order tensor with representation $[\mathbf{S}]$ and $[\mathbf{S}]'$ in $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$, respectively. Show that

$$[\mathbf{S}]' = [\mathbf{A}]^T [\mathbf{S}] [\mathbf{A}].$$

- 1.14 Consider a vector $\mathbf{a}$ with representation $[\mathbf{a}] = (1, 1, 1)^T$ in a coordinate frame $\{\mathbf{e}_i\}$. If $\mathbf{S}$ is an anti-clockwise rotation through an angle of $\pi/4$ about $\mathbf{a}$, find $[\mathbf{S}]$.

- 1.15 For an arbitrary second-order tensor $\mathbf{A} = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$ show that

$$\det \mathbf{A} = \frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} A_{ip} A_{jq} A_{kr},$$

and deduce that $\det \mathbf{A} = \det \mathbf{A}^T$.

- 1.16 For any two second-order tensors $\mathbf{A}$ and $\mathbf{B}$ show that

$$\det(\mathbf{AB}) = (\det \mathbf{A})(\det \mathbf{B}).$$

Moreover, if $\mathbf{A}^{-1}$ exists, show that

$$\det \mathbf{A}^{-1} = 1 / \det \mathbf{A}.$$

- 1.17 For any pair of vectors $\mathbf{u}$ and $\mathbf{v}$ and any invertible second-order tensor $\mathbf{F}$ show that

$$(\mathbf{F}\mathbf{u}) \times (\mathbf{F}\mathbf{v}) = (\det \mathbf{F})\mathbf{F}^{-T}(\mathbf{u} \times \mathbf{v}).$$

- 1.18 Prove Result 1.1.

- 1.19 Show that:

- (a) $|\det \mathbf{Q}| = 1$ for any orthogonal tensor $\mathbf{Q}$,
- (b) $\det \mathbf{Q} = 1$ for any rotation tensor $\mathbf{Q}$.

- 1.20 Prove Result 1.6.

- 1.21 Let $\mathbf{Q}$ be a rotation tensor and let $\mathbf{u}$, $\mathbf{v}$ be arbitrary vectors. Show that:

- (a) $(\mathbf{Q}\mathbf{u}) \cdot (\mathbf{Q}\mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$,
- (b) $|\mathbf{Q}\mathbf{v}| = |\mathbf{v}|$,
- (c) $(\mathbf{Q}\mathbf{u}) \times (\mathbf{Q}\mathbf{v}) = \mathbf{Q}(\mathbf{u} \times \mathbf{v})$.

Remark: The results in (a) and (b) together imply that the length of a vector and the angle between any two vectors are unchanged by a rotation. The result in (c) implies that rotations commute with the cross product operation; in particular, when two vectors are subject to a common rotation, the normal to their plane is subject to the same rotation.

- 1.22 Let $\mathbf{Q} \neq \mathbf{I}$ be a rotation tensor.

(a) Show that $\lambda = 1$ is always an eigenvalue of $\mathbf{Q}$. Hint: Use the characteristic polynomial and properties of determinants.

(b) Show that there is only one independent eigenvector $\mathbf{e}$ such that $\mathbf{Q}\mathbf{e} = \mathbf{e}$. Hint: Use part (c) of Exercise 21 to show that if there were more than one such independent eigenvector, then there must be three, which would imply $\mathbf{Q} = \mathbf{I}$.

(c) Let $\mathbf{n}$ be any unit vector orthogonal to $\mathbf{e}$. Show that $\mathbf{Q}\mathbf{n}$ is also a unit vector orthogonal to $\mathbf{e}$ and that the angle $\theta \in [0, \pi]$ between $\mathbf{n}$ and $\mathbf{Q}\mathbf{n}$ satisfies the relation

$$1 + 2 \cos \theta = \text{tr } \mathbf{Q}.$$

Hint: Express $\mathbf{Q}$ in the frame $\{\mathbf{e}, \mathbf{n}, \mathbf{e} \times \mathbf{n}\}$.

Remark: The vector $\mathbf{e}$ in part (b) is called the **rotation axis**

and the angle θ in part (c) is called the **rotation angle** of $\mathbf{Q}$. The action of $\mathbf{Q}$ on any vector $\mathbf{v}$ can be understood by decomposing $\mathbf{v}$ into parts parallel and perpendicular to $\mathbf{e}$.

- 1.23 Show that the principal invariants of a symmetric second-order tensor $\mathbf{S}$ are given by

$$\begin{aligned} I_1(\mathbf{S}) &= \lambda_1 + \lambda_2 + \lambda_3, \\ I_2(\mathbf{S}) &= \lambda_1 \lambda_2 + \lambda_1 \lambda_3 + \lambda_2 \lambda_3, \\ I_3(\mathbf{S}) &= \lambda_1 \lambda_2 \lambda_3, \end{aligned}$$

where λ_i are the eigenvalues of $\mathbf{S}$. Hint: Choose a simple frame in which to represent $\mathbf{S}$.

- 1.24 Let $\mathbf{S}$ be a second-order tensor and let $I_2(\mathbf{S})$ be its second principal invariant. Show that $I_2(\mathbf{S})$ has the same numerical value regardless of the coordinate frame in which it is computed.

- 1.25 Prove Result 1.10 for symmetric $\mathbf{S}$.

- 1.26 Prove Result 1.11.

- 1.27 For any vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ show

$$(\mathbf{a} \otimes \mathbf{b}) : (\mathbf{c} \otimes \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}).$$

- 1.28 For any given coordinate frame $\{\mathbf{e}_i\}$ show that the dyadic basis $\{\mathbf{e}_i \otimes \mathbf{e}_j\}$ for $\mathcal{V}^2$ is orthonormal in the inner product (1.17).

- 1.29 Prove Result 1.13.

- 1.30 Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ be second-order tensors. Show that

$$\mathbf{A} : \mathbf{BC} = \mathbf{AC}^T : \mathbf{B} = \mathbf{B}^T \mathbf{A} : \mathbf{C}.$$

- 1.31 Let $\mathbf{a}$ and $\mathbf{b}$ be vectors and let $\mathbf{A}$ be a second-order tensor. Define $\mathbf{C}$ to be the fourth-order tensor given by $\mathbf{C}(\mathbf{S}) = \mathbf{AS}$. Show that, if $\mathbf{a}$ is an eigenvector of $\mathbf{A}$ with eigenvalue α , then

$$(\mathbf{C}(\mathbf{a} \otimes \mathbf{b}))\mathbf{v} = \alpha(\mathbf{b} \cdot \mathbf{v})\mathbf{a}.$$

- 1.32 Find the components C_{ijkl} of the fourth-order tensor defined by

$$\mathbf{C}(\mathbf{S}) = \frac{1}{2}(\mathbf{S} - \mathbf{S}^T).$$

- 1.33 Prove Result 1.14.